

Honeywell

TotalPlant Solution (TPS) System

TPS Fieldbus Overview

FB27-630

R630

12/01

 **TotalPlant**

Notices and Trademarks

**Copyright 2001 by Honeywell International Inc.
Release R630 December 17, 2001**

While this information is presented in good faith and believed to be accurate, Honeywell disclaims the implied warranties of merchantability and fitness for a particular purpose and makes no express warranties except as may be stated in its written agreement with and for its customers.

In no event is Honeywell liable to anyone for any indirect, special or consequential damages. The information and specifications in this document are subject to change without notice.

Honeywell and **TotalPlant** are registered trademarks of Honeywell International Inc.

Other brand or product names are trademarks of their respective owners.

Honeywell
Industrial Automation and Control
16404 North Black Canyon Highway
Phoenix, AZ 85053
1-800 343-0228

About This Document

This document describes,

- Integrating Fieldbus in TPS
- Blocks, objects, and alarms in user applications
- Fieldbus IOP

Release Information

Document Name	Document ID	Release Number	Publication Date
TPS Fieldbus Overview - FB27	FB27-630	R630	12/01

Contacts

World Wide Web

The following lists Honeywell's World Wide Web sites that will be of interest to our industrial automation and control customers.

Honeywell Organization	WWW Address (URL)
Corporate	http://www.honeywell.com
Industrial Automation and Control	http://www.iac.honeywell.com
International	http://content.honeywell.com/global

Telephone

Contact us by telephone at the numbers listed below.

	Organization	Phone Number	
United States and Canada	Honeywell Inc. Industrial Automation and Control	1-800-343-0228	Sales
		1-800-525-7439	Service
		1-800-423-9883	Tech. Support
Asia Pacific	Honeywell Asia Pacific Inc. Hong Kong	(852) 28298298	
Europe	Honeywell PACE Brussels, Belgium	[32-2] 728-2111	
Latin America	Honeywell Inc. Sunrise, Florida U.S.A.	(954) 845-2600	

Symbol Definitions

The following table lists those symbols used in this document to denote certain conditions.

Symbol	Definition
	ATTENTION: Identifies information that requires special consideration.
	TIP: Identifies advice or hints for the user, often in terms of performing a task.
	REFERENCE -EXTERNAL: Identifies an additional source of information outside of the bookset.
	REFERENCE - INTERNAL: Identifies an additional source of information within the bookset.

About This Document
Symbol Definitions

Contents

1.	INTRODUCTION TO TPS FIELDBUS.....	15
1.1	FOUNDATION Fieldbus Fundamentals.....	15
	Introduction to FOUNDATION Fieldbus	15
	Additional resources.....	15
	FOUNDATION Fieldbus Relationship to TPS	15
1.2	TPS Fieldbus Implementation Requirements.....	16
	Hardware platform requirements.....	16
	Software platform requirements	16
2.	INTEGRATING FIELDBUS IN TPS.....	17
2.1	Fieldbus Topology	17
	Honeywell's Way is Solution Centric	17
	TPS Fieldbus Illustration	18
2.2	Fieldbus IOP Purpose.....	18
	HPM Interface to Fieldbus.....	18
	Fieldbus Integration into TPS	19
	Fieldbus Blind Record Data Pipe (BRDP)	19
	Fieldbus Network Loading.....	20
2.3	Fieldbus Network Overview	20
	Flexibility of Network	20
2.4	Interoperability and Interchangeability.....	21
	Interoperability.....	21
	What are Device Descriptions?	21
	Interchangeability Defined.....	21
2.5	Fieldbus Communications	21
	Open Systems Model (OSI)	21
	Fieldbus Message Specification (FMS).....	22
	Physical Layer.....	23
	Communications Stack	23
	Scheduled Communication	23
	Unscheduled Communication	24
	Block Based User Application	25
	Communication between H1 Segments	25
3.	BLOCKS IN USER APPLICATIONS.....	27

Contents

3.1	Function Block Application Process (FBAP).....	27
	Function Block Application Process Elements	27
	What is an FBAP?	27
3.2	Block Descriptions	28
	FBAP Elements	28
	Device Objects	28
	Block Objects	28
3.3	Resource Block.....	29
	Each Device has Resource Block	29
	What is a Capabilities File?	29
	Resource Block Information	29
	Basic Block.....	29
3.4	Transducer Blocks	29
	Purpose of Transducer Blocks	29
3.5	Function Blocks.....	30
	Purpose of Function Blocks.....	30
	Custom (Basic) Function Block	30
	Standard FF Function Blocks	30
	Enhanced (Manufacturer-Specific) Blocks	30
	Simulation on Analog Input/Output Blocks	30
3.6	Analog Input Function Block.....	30
	Analog Input Block Purpose	30
3.7	PID Function Block.....	31
	PID Function Block Purpose	31
	PID Control Signals	31
	PID Function Block Schematic	32
3.8	Analog Output Function Block.....	32
	Analog Output Block Behavior.....	32
	Analog Output Block Diagram	32
	Analog Output Block Schematic	33
4.	OBJECTS IN USER APPLICATIONS	35
4.1	Link Objects	35
	Purpose of Link Objects	35
	Linkages	35
	Fieldbus Message Specification	36
	Virtual Communication Relationships (VCR)	36
	Publisher-Subscriber	36
	Client-Server	37
4.2	View Objects	37
	Role of View Objects	37

4.3 Alarm and Alert Objects	37
What are Alerts?	37
Alert Object Purpose	37
Alarms are alerts	37
Configure Alarms from TPS Builder	38
Features and Features Sel.....	39
4.4 Additional Fieldbus Object Support.....	39
Trend Objects.....	39
Domain Objects.....	39
Object Dictionary	40
4.5 Device Descriptions	40
Standardize Device Description	40
TPS Builder templates	40
5. ALARMS IN USER APPLICATIONS	41
5.1 Alert Acknowledgments	41
Two Types of Alert Acknowledgments	41
Implicit Operator Acknowledgment.....	41
5.2 Alarms	43
Alarm Priority Functionality.....	43
Alarm Parameters	44
Time-Tagging	44
5.3 Device and Link Alarms.....	44
Block Alarms	44
5.4 Standard Process Alarms	45
Device Generated Analog Alarms	45
Soft Failures	45
IOP Box Soft Failures.....	46
IOP Generated Slot Soft Failures.....	46
Device Generated Soft Failures	46
Enabling and Disabling Alarms	48
5.5 Suppression of Device Alarms and Process Alarms During a Device Failure	49
Suppression of Device Alarms	49
Suppression of Process Alarms	49

6.	DEVICE MANAGEMENT IN USER APPLICATIONS.....	51
6.1	System Management	51
	System Management VFD	51
	TPS Builder VFD Templates	51
	System Management Key Features	51
	System Application Clock Time Synchronization.....	51
6.2	Device Identification.....	51
6.3	Scheduling a Control Strategy	52
	Function Block Scheduling	52
	Link Active Schedule	52
	Network Management	53
7.	FIELDBUS IOP	55
7.1	Personality and Point Types	55
	Fieldbus IOP Personality.....	55
	HPM Fieldbus Point Types and Slot Numbers	55
	Software Slot Assignment	55
7.2	SECM Point Type	55
	Segment Entity Control Module.....	55
	Command SECM shutdown from Device Detail	57
	SECM Livelist Bitmap	58
7.3	PECM Point Type	58
	Physical Entity Control Module.....	58
	PECM Point Detail Displays	59
	Targets to PECM Pages.....	60
	PECM Resource Block View	60
7.4	FBCM Point Type	61
	Function Block Control Module.....	61
	Block Pages	61
7.5	Parameter Mapping and Projection	62
	Access to Parameters	62
	Mapped Parameters.....	62
	Projected Parameters.....	63
	Structured Parameters	63
7.6	Peer-to-Peer Connections to Field Device Parameters	63
	Field Interconnections	63
	Control Interconnections	63
	Remote Out or Direct Digital Control	63

Consistent Control Relationship	64
Control Connections through HPM.....	64
7.7 Parameter Support in Displays.....	65
Schematics.....	65
Fieldbus Additions to Conventional Displays.....	65
New Displays for Fieldbus.....	65
Calling Fieldbus Network Displays	66
Calling Fieldbus Control Module Displays	67
CL Restrictions.....	68
Group Display	68
Metadata Based Detail Display	68
Detail Display	69
8. HPM AND FIELDBUS CONTROL CONSIDERATIONS	71
8.1 HPM Control Loss, Failure, and IOP Restart.....	71
HPM Control Loss	71
HPM Failure	71
Fieldbus IOP Restart.....	71
8.2 HPM Failover to Fieldbus Control	72
9. DISPLAYING HPM FIELDBUS POINTS.....	73
9.1 Schematic Parameter Access.....	73
Accessibility defined in PRD.....	73
9.2 Group and Detail Display Behavior.....	73
Group Display Behavior	73
Point Type Detail Displays	74
Status Byte Indication	74
9.3 Display Mode Handling.....	75
Mode Management	75
Rotary and Switch Models.....	75
Configuration Options	76
9.4 Status Displays.....	76
Displays overview	76
IOM Detail Display	77
HPM/Fieldbus Checkpointing	77
FB IOP Personality Upgrade	77
Segment Status Display Pages.....	80
Network Statistics Display	81

10. DISPLAY INDICATIONS OF FF STATUS BYTE	83
10.1 Status Byte and Data Structure Overview	83
Status Byte Information	83
Data Structures	83

Figures

Fieldbus in TPS Network	18
Fieldbus IOP in HPM	19
Basic Fieldbus Network with One Device.....	20
FF Device Interaction.....	22
Inter-Device Communication	23
FBAP Block Diagram	27
AI Function Block.....	31
PID Function Block Diagram	31
Link and Alert Objects in Fieldbus	36
Features.....	39
Analog Input Detail Page Showing TPN Alarm Parameters.....	42
Analog Input VIEW 4 Page Showing FF Alarms	43
Fieldbus PID Detail Page showing Alarm Parameters	44
Transducer Block VIEW Page	45
Fieldbus Soft Failures	47
IOP Slot Summary Showing Soft Failures.....	48
Loop 1 Example.....	52
LAS Scheduling for Loop 1	52
SECM Point Detail Display	56
Command targets on SECM Device Detail Display	57
Segment Status Display Page 1	58
PECM Point Display	59
PECM with Targets to View Resource Block Pages	60
Resource Block — Operator Dynamic, Subpage 1	61
FB VIEW Analog Input Operator Dynamic Data.....	62
Regulatory Control in HPM, AI, Backup PID and AO in Field	63
Calling Tree Fieldbus Displays	66
Calling Tree CM Point Detail Displays.....	67
Example of Group Display	68
Example of AI Point Detail Display	69
Input in Fieldbus Device, Regulatory Control and Output in HPM	71
RegCtl PID in HPM, AI, AO and Backup PID in Fieldbus.....	72
Group Display Including Fieldbus Points.....	74
IO Module Display for a Fieldbus IOP	77
HPM Status Display with Fieldbus IOP Personality Reload	78
IOP PI Loading Information Display	79
Fieldbus Segment Status Display.....	80
Fieldbus Network Statistics	81

Contents

1. Introduction to TPS Fieldbus

1.1 FOUNDATION Fieldbus Fundamentals



ATTENTION

You will see the terms FOUNDATION Fieldbus and Fieldbus Foundation used throughout this document. FOUNDATION Fieldbus represents the network technology while Fieldbus Foundation represents the standards organization.

Additionally, "fieldbus," as used in this document also refers to FOUNDATION Fieldbus.

Introduction to FOUNDATION Fieldbus

If you are unfamiliar with the basic fundamentals of FOUNDATION Fieldbus, you should first read the *TPS Fieldbus Fundamentals* document. For an overview, refer to the Introduction, What is Fieldbus? section.

Additional resources

Additional fieldbus literature is available from the Fieldbus Foundation:

- Technical Overview
- FF-003-1.3 Communication & User-Layer Technical Specifications. This is the specification in paper form or CD-ROM.
- Wiring and Installation
- Intrinsic Safety

Information about Fieldbus Foundation is available at their website,
<http://www.fieldbus.org/>

FOUNDATION Fieldbus Relationship to TPS

Honeywell has implemented FOUNDATION Fieldbus (FF) into its TotalPlant Solutions (TPS) system giving you a standard interface to fieldbus as well as other Honeywell hardware and software applications. This implementation is discussed in Fieldbus Topology in the "Integrating Fieldbus in TPS" section.

1.2 TPS Fieldbus Implementation Requirements

Hardware platform requirements

The minimum TPS hardware requirements for fieldbus are:

- High-Performance Process Manager (HPM)
- Fieldbus IOP
- Fieldbus FTA
- IOP/FTA communication and power cables
- Fieldbus network cable
- FOUNDATION Fieldbus certified fieldbus device(s)
- NI Configurator PC interface card
- Global User Station
- Primary Domain Controller (NT) Server
- TotalPlant Network (TPN) Server
- History Module
- Network Interface Module (NIM)

Software platform requirements

Fieldbus requires the following software:

- Honeywell R610
- Global User Station Release 210 or later
- NI Configurator software
- TPS Builder Release 120
- Microsoft Excel or Excel 97.
- Premia Codewright Professional 5.1
- OLE Automation controller (such as Microsoft Visual Basic 5.0 or Microsoft Excel 7.0 97)
- Microsoft Access 7.0 (or other ODBC compliant application).

2. Integrating Fieldbus in TPS

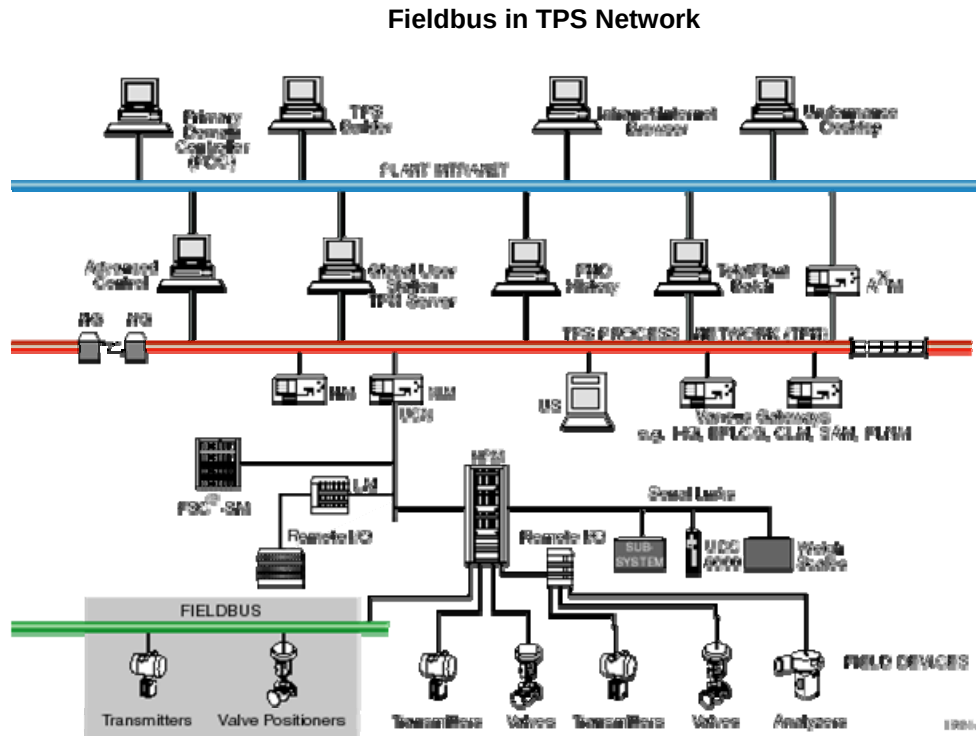
2.1 Fieldbus Topology

Honeywell's Way is Solution Centric

Fieldbus is an enabling technology in Honeywell's offering of **TotalPlant** open solutions. Honeywell has taken a "solution-centric" rather than a "field centric" approach. A field-centric approach, promoted by some suppliers, implies that field instruments and final control elements connected to personal computers will be sufficient to serve users' needs. The solution-centric approach to fieldbus unifies the elements that drive production and presents them in meaningful ways to engineers and operators who run the plant, to maintenance personnel, and to business information systems.

TPS Fieldbus Illustration

The following figure illustrates the integration of fieldbus into TPS.



2.2 Fieldbus IOP Purpose

HPM Interface to Fieldbus

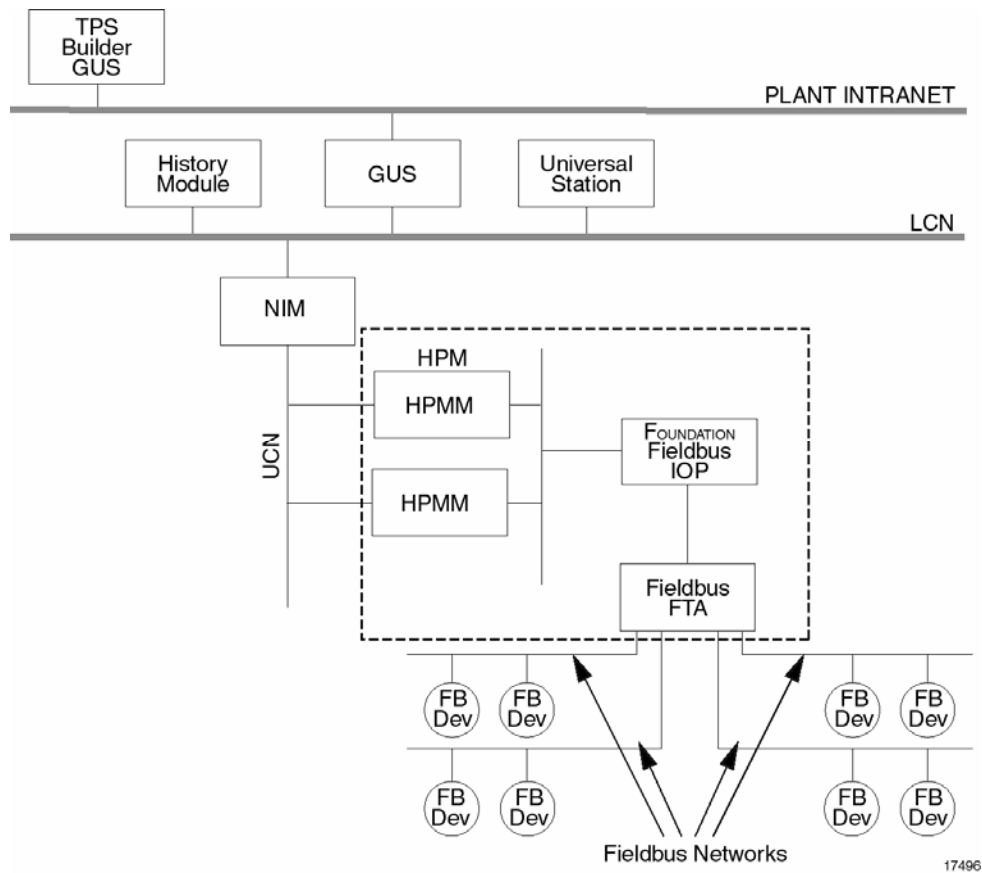
Fieldbus interfaces to the TotalPlant system through a special IOP card which resides in the HPM. Each fieldbus IOP has a corresponding Field Termination Assembly (FTA) to which up to four fieldbus H1 segments may be connected. This form of implementation allows fieldbus data to flow along the fieldbus to the connected devices and through the IOP to TPS operating displays, business systems, and historization functions.

Fieldbus networks and interconnecting devices are configured using TPS Builder, which documents and stores the configuration data. Data is downloaded to fieldbus devices through the LCN, NIM, UCN, HPM and fieldbus IOP.

Fieldbus Integration into TPS

A new IOP and FTA supports fieldbus integration into the HPM, as shown in the following figure.

Fieldbus IOP in HPM



Fieldbus Blind Record Data Pipe (BRDP)

The BRDP is the mechanism by which all information, including configuration and parameter data, is passed upward and downward through the IOP to and from LCN UCN, NIM, etc. This implies that the data need not be “understood” by the network(s) through which it is passing and that the sender/receiver of the information can understand it.

Fieldbus Network Loading

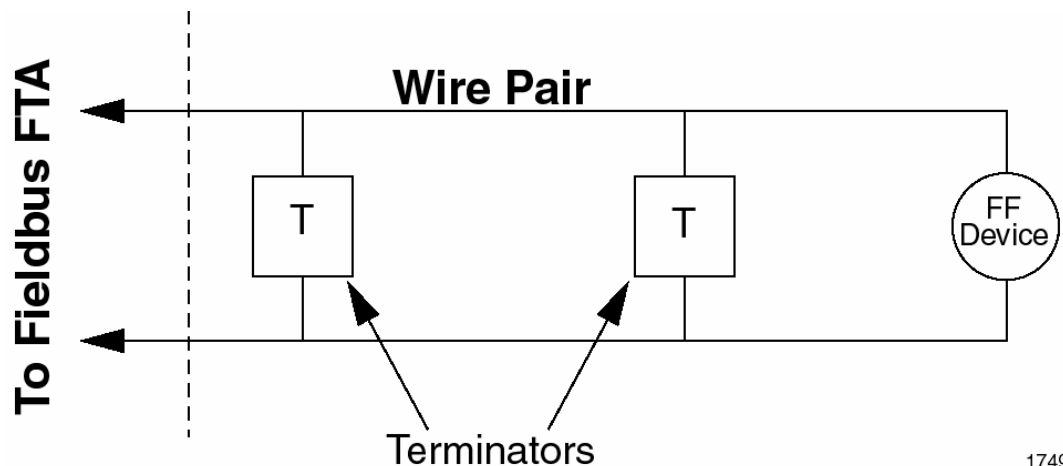
The Fieldbus IOP (FB-IOP) is capable of supporting four H1 segments and 119 control modules. These control modules represent physical devices, function blocks, and transducer blocks. A single function block control module can represent a function block and a transducer block, if one is present. A fieldbus device is represented by a single physical entity control module. Each fieldbus segment can support up to 10 fieldbus devices. Refer to the Fieldbus Planning, Installation, and Servicemanual for guides to cable lengths, and spur lengths.

2.3 Fieldbus Network Overview

Flexibility of Network

Fieldbus networks are extremely flexible and can be simple or extremely complex. We recommend that you become familiar with the different topologies in the *Fieldbus Planning, Installation, and Service* manual. Fieldbus networks consist of a twisted-wire pair, and as implemented in the TPS system are connected to a Fieldbus Foundation Interface (FFI). The interface for H1 field bus is a special IOP in the HPM. Each fieldbus IOP has a corresponding Field Termination Assembly (FTA) to which up to four fieldbus H1 segments may be connected. A fieldbus network can be constructed using an analog wire pair, but communication wire is recommended. An example of a simple fieldbus using existing 4-20-mA wire is shown in the following figure.

Basic Fieldbus Network with One Device



17492

2.4 Interoperability and Interchangeability

Interoperability

To ensure that devices connected to a fieldbus can share information, Device Description technology and Capabilities Files are used. The manufacturer-supplied Capabilities File is consistent with FF standards.

The key to devices' ability to be understood by configuration tools and by a host system are the device descriptions (DD). DDs define the resource, function, and transducer blocks contain descriptions of parameters that exist on blocks. FF certified devices contain the physical device, interface, FF communication stack, function blocks and parameters.

What are Device Descriptions?

DDs contain standard operational, control, and engineering information, plus configuration and maintenance data. They may also contain manufacturer-specific function block and transducer block parameters. Manufacturers may also supply Custom Blocks not derived from an FF standard.

TPS Builder uses the DDs and the Capabilities File to build templates which are used by the builder function to create strategies and build points associated with the strategies.

Interchangeability Defined

Devices are not interchangeable *per se*; that is, when a device is replaced with a similar or equivalent device, the old device must be deleted from the database and the new one is configured through TPS builder.

Devices may be replaced with *identical devices*, providing the replacement device is identical in every respect, including the revision number, except the serial number. The database must be edited in TPS Builder for a nearly equivalent device, and any device mismatches must be corrected.

The original configuration data is obtained from Capabilities Files, DDs and value files (when provided) and downloaded to a device.

To ensure that each device is unique, it is identified by a unique address on the link. It is also identified by its serial number.

2.5 Fieldbus Communications

Open Systems Model (OSI)

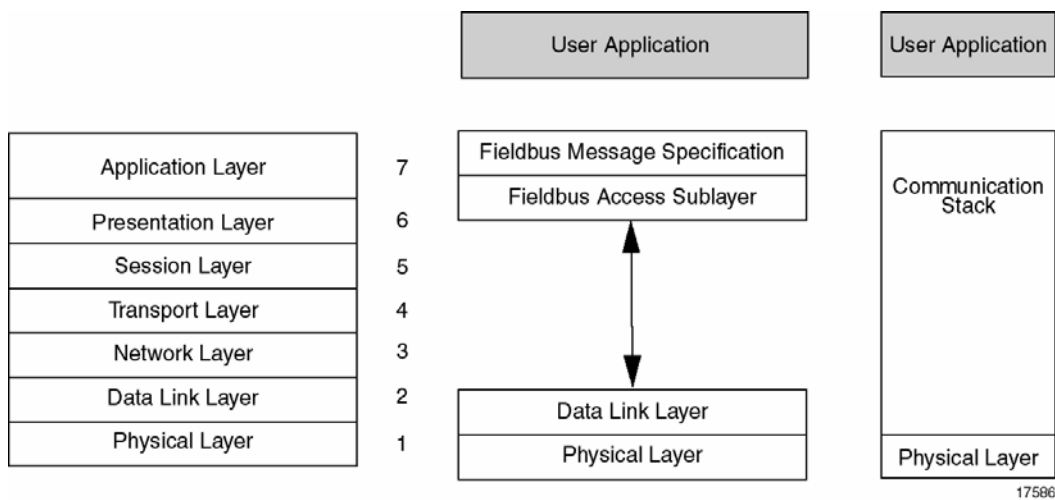
FOUNDATION Fieldbus communications technology consists of three layers:

- Physical layer
- Data Link Layer
- Application Layer

Additionally, there is a User Layer which is largely based on function blocks.

A subset of the ISO's OSI reference model is used to model these layers, as shown in the following figure. The physical layer is layer 1; the data link layer is layer 2; the Fieldbus Message Specification (FMS) is layer 7 (layers 3 through 6 are not used). Fieldbus does not use layers 3 through 6.

FF Device Interaction

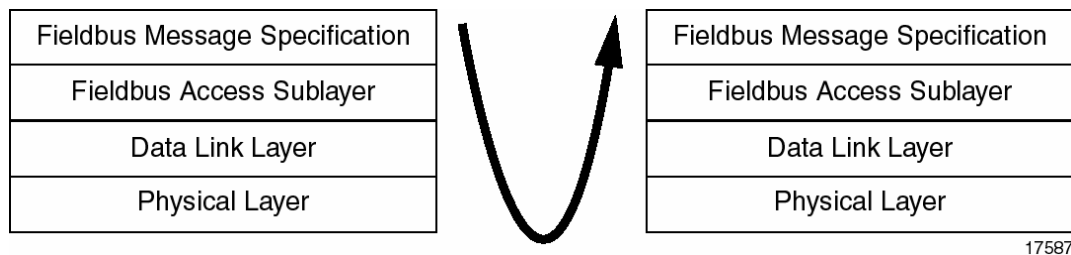


Fieldbus Message Specification (FMS)

FMS services, in combination with function block definitions, specify the functional interface for field devices, specify the message format, and the function block definitions define the data contained in the messages.

FMS allows different devices and the host system to communicate with each other across the fieldbus using FF formats. It describes the different services, formats and protocol needed to build messages for the User Application. FMS controls the level of access for these messages and also records the access privilege of the users. Intercommunication between devices on the FB network is shown in the following figure.

Inter-Device Communication



Physical Layer

The physical layer is defined by approved standards from the IEC and the ISA. It receives the messages from the communications stack and converts them into signals for transmission. It is a synchronous serial because clock information is embedded in the serial data stream.

Communications Stack

The communications stack shown in an earlier figure is composed of the Data Link Layer (DLL), Fieldbus Access Sublayer, and Fieldbus Message Specification. The Data Link Layer (2), transmits messages on the fieldbus through the Physical Layer (1). The DLL manages access to fieldbus through a “deterministic” centralized bus scheduler, the Link Active Scheduler (LAS). The LAS controls the exchange of data between devices and schedules transmission of deterministic messages. The DLL defines two types of devices:

- Basic Device
- Link Master

Each H1 has a device which is the Link Master and optionally one or more devices which are capable of assuming the role of backup to the Link Master. Under normal operating conditions the Fieldbus IOP serves as the master of the fieldbus segment. The backup Link Master is designated during configuration and can be any device on the fieldbus containing algorithms. A device such as the Honeywell ST3000 can assume this role when configured as the backup Link Master.

Scheduled Communication

Scheduled communication on a Fieldbus network is supported through the use of

- Link Master,
- Link Active Scheduler, and

- Compel Data message.

Link Master

A Link Master is a class of device which can contain a Link Active Schedule. In the TPS system the Fieldbus IOP is configured as the Primary Link Master.

Link Active Scheduler

The fieldbus network is designed for time-critical applications. For fieldbus synchronous control, data updates are pre-scheduled to occur at specified times under the control of a Link Active Scheduler (LAS). These updates are broadcast by the “publisher” device which is the source of the data, and are received by all “subscriber” devices. The data is accompanied by data quality and quality information. The field devices have a shared sense of time, which can be synchronized to within less than a millisecond (the smallest time unit granule is 1/32 of a millisecond on the fieldbus), to support device internal scheduling operation.

The LAS broadcasts a Time Distribution so that all devices on the data link have exactly the same link time. This facilitates the prompt scheduling of all LAS activities.

Compel Data (CD)

The Compel Data function is a list of schedules by which *cyclic* activities occur. At the time scheduled, the LAS sends a CD message to a specific message buffer in a specific device; the device immediately publishes the data to all devices on the fieldbus. The CD is given highest priority by the LAS; other actions are performed between scheduled transfers.

Unscheduled Communication

Unscheduled communication on a Fieldbus network is supported through the use of

- Token passing message,
- Link access delegation, and
- Live List maintenance.

Token Passing Message

The LAS gives permission to devices to transmit messages on the network by issuing a Pass Token (PT) message to the device. When the device receives the token, it can transmit messages until it has finished or until its “hold time” expires, whichever is shorter.

Link Access Delegation

All devices on the fieldbus network can send an “unscheduled” message through the Pass Token (PT) mechanism. The LAS supports unscheduled traffic by delegating link access in turn to each device. The scheduling and token delegation mechanism guarantees that scheduled data transfers will occur on time and that the first high priority message from a device will be sent within a specified time.

Live List Maintenance

Devices that currently respond to the Pass Token (PT) are kept in a list known as the **Live List**. Periodically, the LAS issues a “probe node” to addresses which do not appear in the live list. If the device is present at the address, it issues a response and the LAS adds it to the live list. After completing the cycle of sending PTs to all devices in the list, the LAS issues a Probe Node to at least one address.

Block Based User Application

The user application is based on FF “blocks.” Block types are:

- **Resource blocks**, identifying the characteristics of the device
- **Function blocks**, defining the system behavior such as; input (AI, DI), output (AO, DO), or control (PID)
- **Transducer blocks**, containing information such as type of sensor, calibration date

Communication between H1 Segments

Communication between H1 segments is made through HPM resident points, such as a Regulatory Control PID point.

3. Blocks in User Applications

3.1 Function Block Application Process (FBAP)

Function Block Application Process Elements

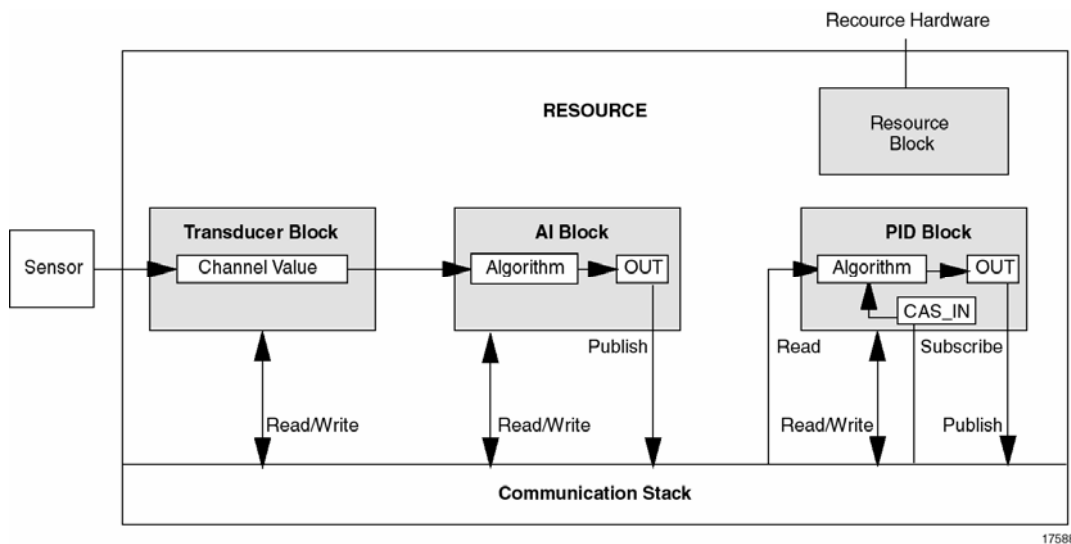
The Function Block Application Process (FBAP) or User Application uses resource blocks, transducer blocks, system and network management functions, and device descriptions. These elements are combined to perform I/O for automated monitoring and control, alarming, and communication within the fieldbus network and the TPS system. This section contains the following topics:

What is an FBAP?

The Function Block Application Process (FBAP) comprises a set of elementary functions that are modeled as “blocks.” These blocks provide a general structure for defining types of device functions, links, communication and transfer of data.

A block diagram shown in the following figure represents some of the key elements involved in the FBAP and their interaction.

FBAP Block Diagram



3.2 Block Descriptions

FBAP Elements

FOUNDATION Fieldbus refers to the functions as “objects.” The key elements of the FBAP are device objects that are:

- Blocks objects and their parameters, which consist of the following block types:
 - Resource blocks
 - Transducer blocks
 - Function blocks (FB)
- Link objects
- Alert objects
- View objects

Device Objects

The most frequent use of Link objects is to specify I/O parameter linkages between function blocks. All blocks may use a link object to specify an alert notify or trend report that is to be transmitted an alarming or reporting device. A brief description of these objects follows.

Block Objects

Blocks are some of the key elements in the FBAP. Each fieldbus device has a resource block, one or more function blocks, and optionally a transducer block. The resource block, derived from the DDs and Capabilities File, represents the physical device. Function blocks provide the control behavior (for example, a PID controller, analog input, etc.) Transducer blocks represent an interface to sensors such as temperature, pressure, flow, and actuators that provide digital and analog output. These blocks are software units that represent functions, data, parameters, and connections for interoperability between blocks.

Configuration of the blocks is accomplished through a TPS Builder Project application, using a block strategy that represents the actual data in the devices.

3.3 Resource Block

Each Device has Resource Block

Each FBAP within a device must have one resource block that describes the physical characteristics of the device. A device may have several FBAPs. The resource block contains FF standard parameters and alert reporting, and may contain manufacturer-specific block extensions, and manufacturer-specific alerts. The device manufacturer generally provides the default block information in a Capabilities File with the device.

What is a Capabilities File?

The Capabilities File contains any parameters in the device, including the resource block, transducer blocks, function blocks and System Management data. TPS Builder uses the DDs and the Capabilities File to create TPS Builder templates.

Resource Block Information

Examples of information contained within the resource block are:

- Name of device.
- Manufacturer.
- Revision level.
- Alerts.
- Status information such as modes, O/S (out-of-service), Auto, and IMan.
- Control timers for high-level Remote Cascade and Output control.

Basic Block

TPS Builder provides templates for the basic block. All FF blocks must support the first six parameters and a block header. Manufacturers may create a “Custom Block” that contains the header, the first six parameters, and manufacturer-specific parameters as defined in the DD. TPS Builder supports this type of block.

3.4 Transducer Blocks

Purpose of Transducer Blocks

Transducer blocks de-couple (or insulate) function blocks from local I/O devices, such as sensors, actuators. Transducer blocks may be of three types: Fieldbus Foundation standard, Fieldbus Foundation Standard Enhanced, or (Custom) Manufacturer-specific. Manufacturer-specific blocks (or Custom blocks) can be derived from a standard block or

a generic block. The TPS block library contains templates for standard and manufacturer-specific transducer blocks.

3.5 Function Blocks

Purpose of Function Blocks

Function blocks describe the control system behavior and may be linked to other blocks; an FBAP can contain many function blocks. The function block inputs and outputs may be linked strategically to other blocks or to control algorithms resident in the HPM. Function blocks are actually software “objects” which connect to other entities in a *standard* manner.

Custom (Basic) Function Block

The custom function block can be used as the basis for a resource, function, or transducer block; it contains a header and the first six required parameters.

Standard FF Function Blocks

These standard blocks have a set of FF parameters, and may have additional manufacturer-specific function block parameters.

Enhanced (Manufacturer-Specific) Blocks

Manufacturers may supply their own enhanced version of a resource, function, or transducer block.

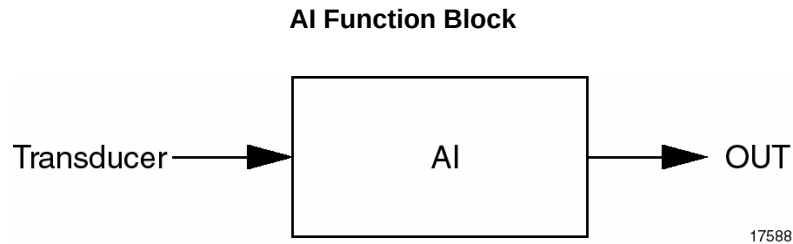
Simulation on Analog Input/Output Blocks

Simulation is a capability of FF that should be used cautiously. When simulation is enabled and active, a value and a status can be “faked” that look real on displays, trends, historians, etc. Input values can look good when they are indeed not real. End-elements can appear to be positioning when indeed they are not. That is intentional in order to be able to test reactions to various values and states of interest.

3.6 Analog Input Function Block

Analog Input Block Purpose

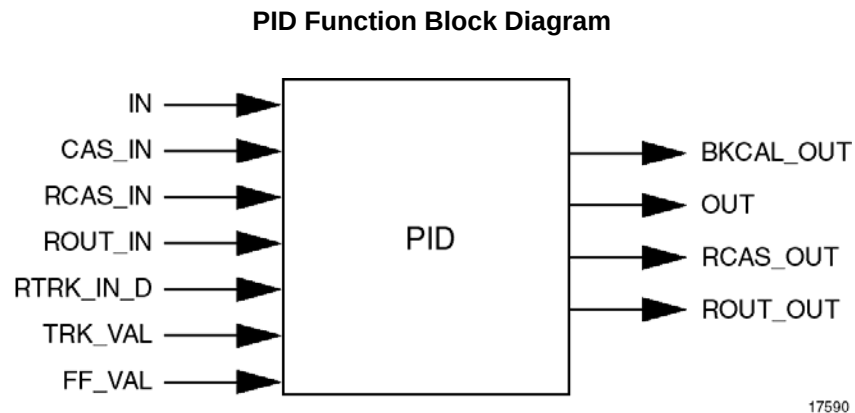
The purpose of input blocks is to access physical measurements through the input transducer block. The block processes the inputs and provides an output for linking to other blocks. Input blocks are composed of the contained parameters; process variable, and output parameters; additional input, output and manufacturer-specific parameters may be defined for the block.



3.7 PID Function Block

PID Function Block Purpose

The PID block is universally used in control strategies because of its ability to integrate errors when they exist, and to move the output in the direction of the error. The PID block may be cascaded to compensate for difference in process time of a primary and secondary measurement, if desired.

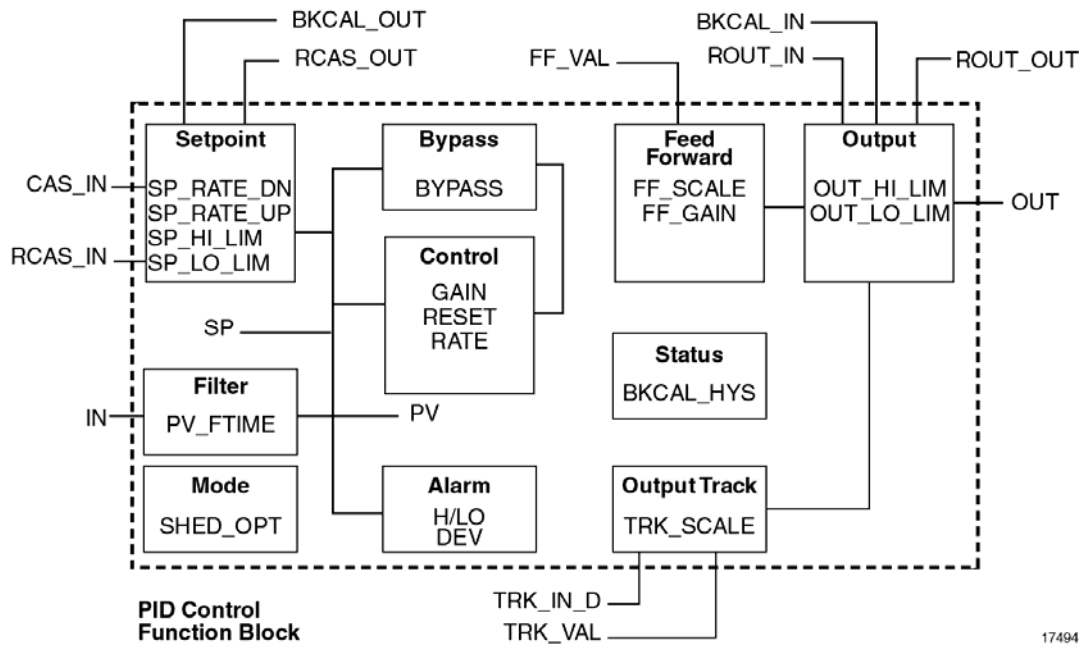


Some manufacturers supply an enhanced version of the PID function block in addition to the FF PID function block. The manufacturer's block may provide several equation types; however the input and output parameters remain the same as the FF block.

PID Control Signals

The PID Control Function produces an output signal in response to a measured variable and a setpoint. The output has three terms, Proportional, Integral and Derivative. The output is adjusted by tuning constants:

PID Function Block Schematic

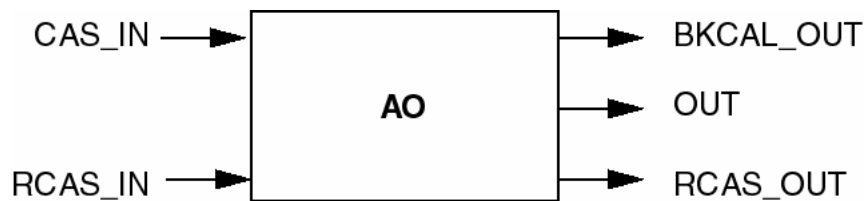


3.8 Analog Output Function Block

Analog Output Block Behavior

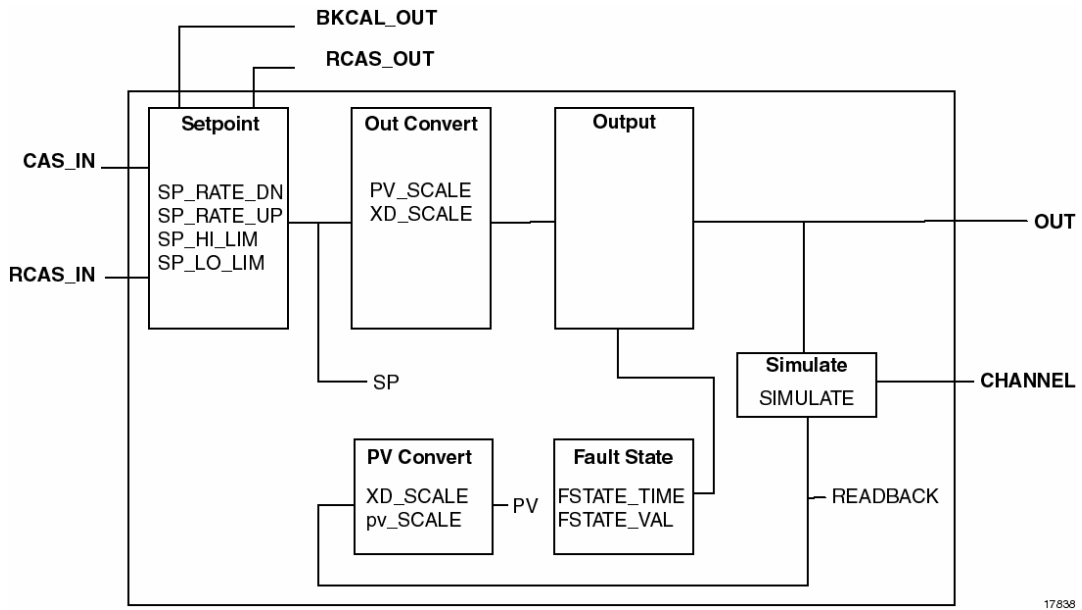
The Analog Output (AO) block, while in cascade mode (Cas or RCas), receives the output (usually from a control block) into one of the cascade_in parameters, **CAS_IN/RCAS_IN**, and provides it to the Setpoint.

Analog Output Block Diagram



17837

Analog Output Block Schematic



4. Objects in User Applications

4.1 Link Objects

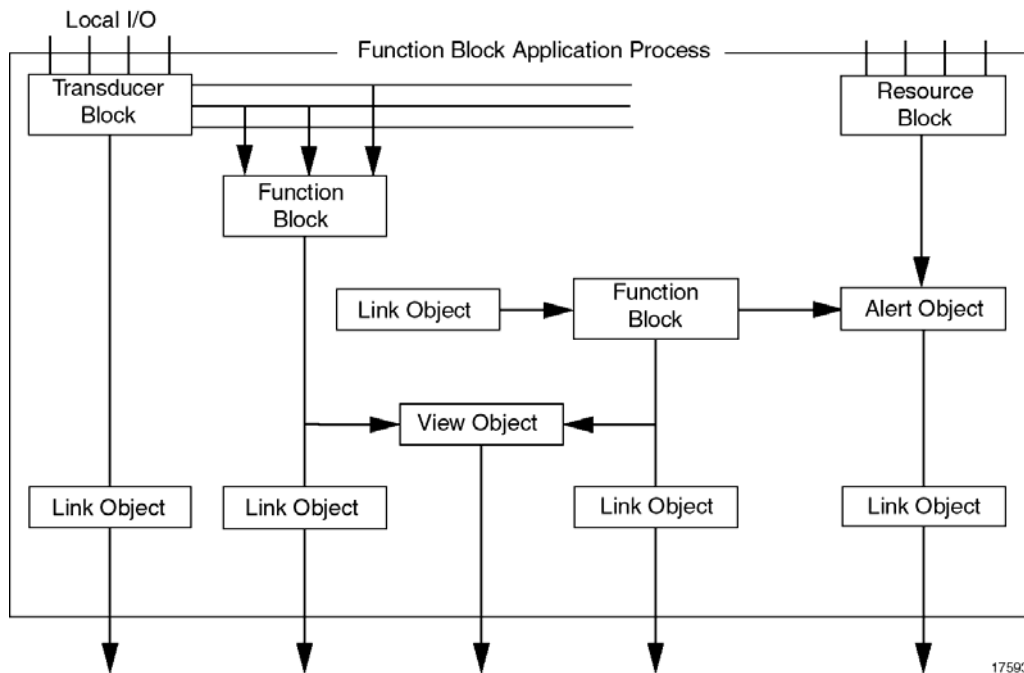
Purpose of Link Objects

Function blocks used to monitor and control a process are linked by “objects” within the devices. These link objects are the method by which data, status, and events (such as alarms) are shared along the fieldbus. These links are defined through link objects. Link objects identify the parameter or object to be linked, the Fieldbus Message Specification (FMS) used to transfer the data and the Virtual Communication Relationship (VCR) used in the transfer.

Linkages

The following figure illustrates the relationship of these linkages within the Function Block Application.

Link and Alert Objects in Fieldbus



Fieldbus Message Specification

User applications communicate over the fieldbus through the FMS services. The FMS describes the format and protocol behavior needed to build messages for the user application.

Virtual Communication Relationships (VCR)

To support the transfer of scheduled and unscheduled message information across the fieldbus network, Virtual Communication Relationships (VCRs) are used. Three types of VCRs are used by the FMS: Publisher/Subscriber, Client/Server, and Report Distribution.

Publisher-Subscriber

The Publisher/Subscriber type VCR is used primarily by field devices for transfer of function block input/output data such as process variables and outputs. The Publisher/Subscriber type VCR is used for one-to-many *buffered* communications. **Only the latest data remains in the buffer.**

Client-Server

When a device on the fieldbus receives the token, it may send a request for a message from another device. The requestor is the *client*; the receiver is the *server*. The server responds when it receives the token. The Client/Server type VCR is used for such transfers as operator-initiated requests, for example setpoint changes, acknowledging alarms, uploading and downloading to devices.

Messages are queued, meaning that messages are sent and received in the order submitted *without overwriting previous messages*.

4.2 View Objects

Role of View Objects

View objects are defined for the resource block, transducer block, and function blocks. The variables in these blocks are mapped into “views.” Some parameters are visible in all four views for display purposes; others may be visible in only one or two views.

4.3 Alarm and Alert Objects

What are Alerts?

In FOUNDATION Fieldbus, events and alarms are known collectively as *alerts*. FF compliant devices must be able to generate device-related and process-related alarms and events. An “alert notifier” (within the resource) reports the detection of an alert condition. An “alert object” communicates this detection to other devices. In this manner, the interface device is relieved of the responsibility for polling the devices for alarming purposes.

Alert Object Purpose

Alert objects are used to generate alarms and report events; this is either an alarm, which has an accompanying return to normal, or an event which is simply the notification of an interesting condition for which there is no return to normal.

Alarms are alerts

Alarms are considered alerts which must be acknowledged by an operator; events are alerts such as soft failures or events which may (or may not) return to normal without operator intervention or which have no return-to-normal condition (such as the placing of a device out of service). Some events are abnormal occurrences reported upward in the system until they reach the operator. These events do require operator acknowledgment.

Configure Alarms from TPS Builder

TPS Builder performs all initial FF alert configuration. In the device this is accomplished by downloading the configuration to the device.

Following is an example of a typical configuration form, the Acknowledge Option (ACK_OPTION) configuration form for an ST3000 AI block.

The screenshot displays the 'Parameters - AI_ST_21' window with the 'Ack_Option' tab selected. The left pane shows various parameters for the AI block, and the right pane shows the 'Ack_Option' configuration options.

Parameters - AI_ST_21

BlockAssign	BlockHead	Parameters-AI	Mfr Parameters	AI Mode-QOS	AI Mode-Set
Use Transducer values direct	L_Type	Direct			
Limit for Flow Sensor Input AI	Low_Cut	0.000000			
Time const. of a single expon	PV_Ftime	0.000000			
Raw Value of Field Device (p	Field_Val	Bad-Out-Of-Service Constant			
Update_Evt	Update_Evt	Acknowledged, Undefined, I			
Used for all Cfg, Hdw, Conn f	Block_Alm	Acknowledged, Uninitialized,			
Current Alert Status and other	Alarm_Sum	0,0,0,0			
Automatically acknowledge A	Ack_Option	0xFFFF			
Amt PV must return within Alm	Alarm_Hys	0.500000			
High High Alarm Priority	Hi_Hi_Pri	0			
High High Alarm setting (in E	Hi_Hi_Lim	+INF			
High Alarm Priority	Hi_Pri	0			
High Alarm setting (in EU)	Hi_Lim	+INF			

Ack_Option

- ☒ Disable/Enable Discrete Alarm
- ☒ Disable/Enable High high Alarm
- ☒ Disable/Enable High Alarm
- ☒ Disable/Enable Low low Alarm
- ☒ Disable/Enable Low Alarm
- ☒ Disable/Enable Deviation high Alarm
- ☒ Disable/Enable Deviation low Alarm
- ☒ Disable/Enable Block Alarm
- ☐ Disable/Enable Alarm-8
- ☐ Disable/Enable Alarm-9
- ☐ Disable/Enable Alarm-10
- ☐ Disable/Enable Alarm-11
- ☐ Disable/Enable Alarm-12
- ☐ Disable/Enable Alarm-13
- ☐ Disable/Enable Alarm-14
- ☐ Disable/Enable Alarm-15

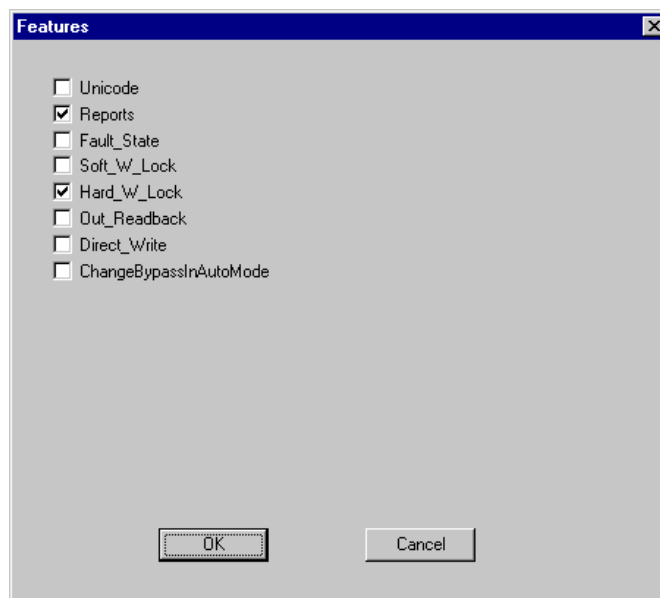
OK Cancel

Features and Features Sel

The manufacturer of a field device has a select group of features that are part of the FOUNDATION Fieldbus specifications; the features provided are defined in the device descriptions. Identification of these features, and the user's options to choose from the provided features are configured in TPS Builder.

Following is an example of a features configuration form.

Features



4.4 Additional Fieldbus Object Support

Trend Objects

Trends for fieldbus values are available in the same manners as other TPN process values by historizing the point and trending the point through the Native Window or Universal station trend functions

Domain Objects

Domain objects are used to support the downloading of the Link Active Schedule to the Management section of a field device.

Object Dictionary

Object dictionaries (OD) reside in the devices to verify the validity of requests to perform FMS services. Object Descriptions describe the data that is communicated across the fieldbus.

4.5 Device Descriptions

Standardize Device Description

Standardized definitions are used to support and describe application process objects. Two of these standardized "tools" used to describe these objects are the Object Dictionary (OD) and the Device Description (DD).

TPS Builder templates

TPS Builder supplies the templates for building the configurations in its various libraries. The user then configures and downloads the device within a project. Configuration and loading of resource and transducer blocks is also done within TPS Builder. Function blocks, and accompanying parameter configuration and tag building, is done through TPS Builder's control strategies. The strategies can then be forward-built and downloaded to individual devices. Forward Build creates an EB file for the selected entities. A new function, Load, is added to all TPSN strategies. The Load function automatically loads TPS Builder configuration data to TPSN, creating an EB file for selected entities and then using Command EC files and the EC Trigger to activate the load on TPSN.

5. Alarms in User Applications

5.1 Alert Acknowledgments

Two Types of Alert Acknowledgments

FF has two types of alert acknowledgments:

- Device Acknowledgement
- Operator Acknowledgements

Implicit Operator Acknowledgment

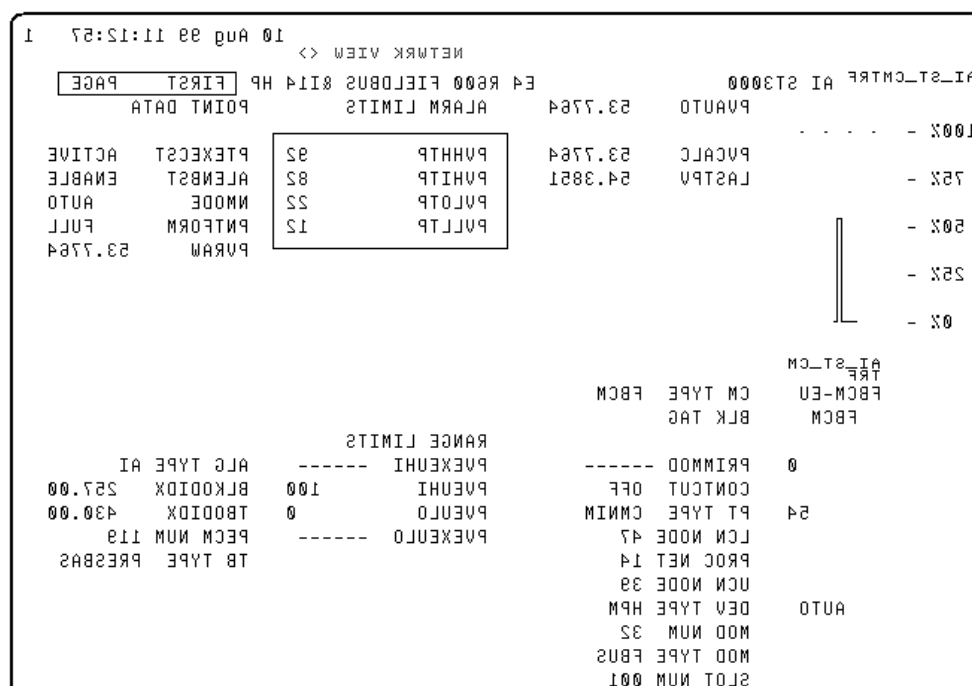
Alarm handling for fieldbus process alarms is the same as for other TPN alarms, as they depend upon configuration parameters for their priorities. Alarms are configured in TPS Builder.

Alarms in User Applications

Alert Acknowledgments

Following is an example of a typical detail page, the AI detail page showing TPN alarms.

Analog Input Detail Page Showing TPN Alarm Parameters



5.2 Alarms

Alarm Priority Functionality

FF events and alarms have been integrated into TPS, including the associated reporting, clearing, journaling and historizing of alarms and events where necessary. Process-related alarms have been mapped to their corresponding TPN parameters.

Alarm limits and priorities may be changed or from the VIEW pages. Following is an example of a typical view page, the AI VIEW 4 page showing FF alarms.

Analog Input VIEW 4 Page Showing FF Alarms

10 Aug 99 13:23:37 1

NETWRK VIEW <>

AI-ST-CMTRF AI ST3000 E4 R600 FIELDBUS 8I14 HP FB VIEW PAGE 4

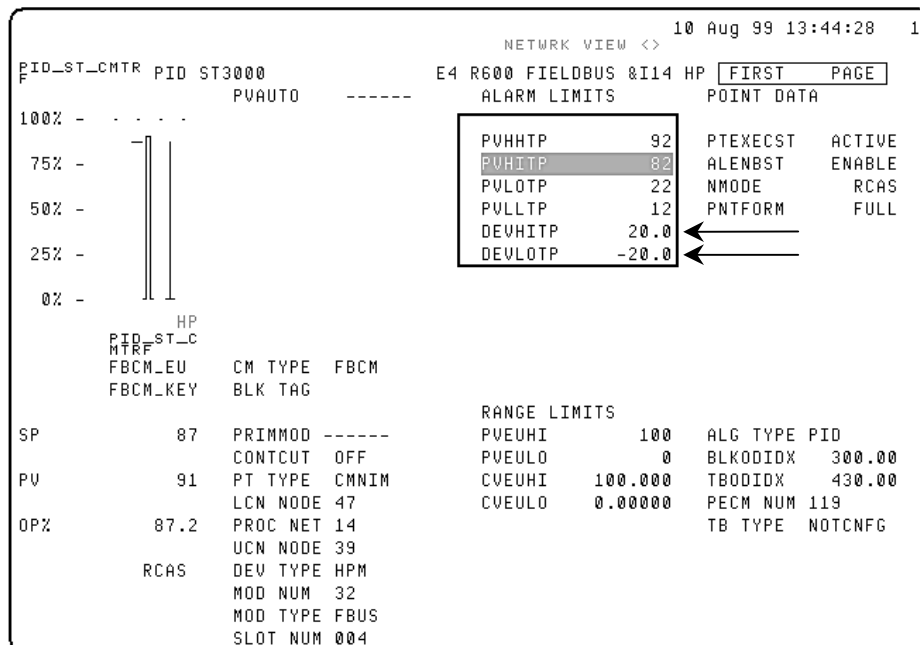
Other Static Data - ST3000 AI Block SUBPAGE 2

Ack_Opti	PARAMETER	INDEX	VALUE
Disable/Enable D	Ack_Option		Disable/Enable Discrete Alarm,
Disable/Enable H			Disable/Enable High high Alarm,
Disable/Enable H			Disable/Enable High Alarm,
Disable/Enable H			Disable/Enable Low low Alarm,
Disable/Enable L			Disable/Enable Low Alarm,
Disable/Enable L			Disable/Enable Deviation high Al,
Disable/Enable D			Disable/Enable Deviation low Ala,
Disable/Enable D			Disable/Enable Block Alarm
Disable/Enable D	Alarm_Hys	0.50000	
Disable/Enable B	Hi_Hi_Pri	15	
Disable/Enable B	Hi_Hi_Lim	92.0000	
	Hi_Pri	12	
MORE	Hi_Lim	82.0000	
	Lo_Pri	12	
	Lo_Lim	22.0000	
ENTER	Lo_Lo_Pri	15	
	Lo_Lo_Lim	12.0000	

Alarm Parameters

Following is an example of a typical fieldbus detail page, the PID detail page showing alarm Parameters.

Fieldbus PID Detail Page showing Alarm Parameters



Time-Tagging

Alerts that are mapped to TPN soft failures and alarms are time-tagged by the HPM in the same way as other IOPs are timestamped.

5.3 Device and Link Alarms

Block Alarms

Block alarms received by the fieldbus IOP are reported to the TPS Network as IO Module soft failures. Any block (function, resource, or transducer) can generate block alarms. Block errors are reported on the VIEW pages with which they are associated.

Following is an example of a typical block page, the transducer block VIEW page showing a block error.

Transducer Block VIEW Page

04 May 99 13:18:24 4

CV_701F
DESCRIPTOR A1 APM 241
TB VIEW PAGE 1

Operator Dynamic Data - 1x_xd_block

SUBPAGE 1

PARAMETER	INDEX	VALUE
ST_Rev		23
Mode_Blk	TARGET	: Auto
	ACTUAL	: OOS
	PERMITTED	: OOS, Auto
	NORMAL	: Auto
Block_Err		Other, OutOfService

Transducer_Type		0
XD_Error		None
Final_Value	50.0000	CONSTANT
Final_Position_Value	0.00000	# DEVICE FAILURE

5.4 Standard Process Alarms

Device Generated Analog Alarms

An analog alarm occurs when a value meets or exceeds a limit. For a high alarm, an alarm occurs when the analog value is greater than or equal to the defined limit. For a low alarm, the directions are reversed. Deviation alarms may be defined where the value tested is the difference between two variables, such as the PV and a designated trip point. FOUNDATION Fieldbus alarms are mapped directly to TPN alarm parameters.

Soft Failures

The Fieldbus IOP Box and Slot Soft Failures are reported on the Network (HPM) Status display on the FBIOP by a flashing red "SF" message.

IOP Box Soft Failures

The following IOP Box Soft Failures apply to fieldbus.

- Bad Personality Image
- Bad Boot Sector and/or Platform Sector Images in Flash ROM
- Bad Cyclic Redundancy Check for Platform Sector Flash ROM Image
- Bad Application Sector Image in Flash ROM
- Stack Hi Water exceeded
- Segment problem or LAS failure
- Segment Diagnostic Overflow
- FF Device Failure
- IOP has just powered up and stacks are still loading

These events are time stamped by the HPM.

IOP Generated Slot Soft Failures

The following IOP Box Soft Failures apply to fieldbus.

- Fieldbus segment has failed
- VFD matching is in progress
- VFD mismatch
- Device not present

Device Generated Soft Failures

Block Alarms received by the fieldbus IOP are reported to TPN as IOM Soft Failures. The first sixteen alarms can be generated by any block (function, resource, or transducer), while the last ten in this table are generated only by transducer blocks.

Fieldbus soft failures

Following is an example of a soft failures page, the fieldbus soft failures page.

Fieldbus Soft Failures

04 May 99 13:58:064

UCN 001HPM 019MODULE 007

Select Page:

12345

678910

Module Display

Box Failures

Slots w/Failures

80 *** Box soft failure only ***

81 Segment Failure

82 VFD Verification

83 VFD Mismatch

84 Device Missing

⇒ 85 FF Block: Other Error

86 FF Block: Block Configuration Error

87 FF Block: Link Configuration Error

⇒ 88 FF Block: Simulation Active

89 FF Block: Local Override

90 FF Block: Device Fail Safe Set

91 FF Block: Device Needs Maintenance Soon

92 FF Block: Input Failure

93 FF Block: Output Failure

94 FF Block: Memory Failure

95 FF Block: Lost Static Data

Slot Soft Failures - Page 6

R630
12/01

TPS Fieldbus Overview
Honeywell

47

IOP slot soft failures

The IOP Slot Summary also indicates the slots with soft failures as shown in the following figure. The "SF" beside the slot number is the indication.

IOP Slot Summary Showing Soft Failures

```

MAKE SELECTION
                                NETWORK VIEW <>
                                10 Aug 99 15:06:10
                                1
                                IOP SLOT SUMMARY
                                UCN 14  DEVICE 39  MODULE 32 - FBUS
                                PAGE 1
                                STS SLOT POINT ID          CMTYPE          DESCRIPTOR          PTEXECST
                                SF      1  AI-ST-CMTRF      FBCM  AI ST3000
                                2  PID-LX-CMTRF      FBCM  LX 1400
                                3  AO-LX-CMTRF      FBCM  AO LX1400
                                4  PID-ST-CMTRF      FBCM  PID ST3000
                                5  RST3K1-AIF      FBCM  REAL ST3000 AI FBCM
                                6
                                7  RST3K1-PIDF      FBCM  PID IN REAL ST3000
                                8
                                9  BPIDF      FBCM  PID ST3K-B IOP32 LINK 2
                                SF  10  BAIF      FBCM  AI ST3K-B IOP32 LINK 2
                                SF  11  AAIF      FBCM  AI ST3K-A IOP32 LINK 2
                                12  APIDF      FBCM  PID ST3K-A IOP32 LINK 2
                                13  BLX-AOF      FBCM  AO LX1400 IOP32 LINK 2
                                14  BLX-PIDF      FBCM  PID LX1400 IOP32 LINK 2
                                15  ST72PIDF      FBCM  10DEV LINK
                                SF  16  ST72AIF      FBCM  10DEV LINK

```

PAGE
SELECT

SLOT
SELECT

PM
STATUS

Enabling and Disabling Alarms

All process alarms except the IOP generated BADPV alarm can be enabled/disabled according to the access level configured for TPN alarms through:

- ALARM_SUM, XXX_PRI, or XXXX LIM on the VIEW data pages, or
- the TPN priority parameter XXXXXPR, or
- the TPN limit parameter XXXXTP.

5.5 Suppression of Device Alarms and Process Alarms During a Device Failure

Suppression of Device Alarms

A user can disable and clear out the existing system soft failures associated with a non-operational FBUS device by setting the respective PECM's point execution state parameter (PTEXECST) to INACTIVE. This prevents nuisance system alarms from being generated during the time that the FBUS device is non-operational.

When the PTEXECST parameter is set to INACTIVE:

- The FBIOP generates return-to-normal events for all outstanding soft failures associated with that device. These include soft failures from it's related PECM and all related FBCMs.
- The generation of any further soft failures for the failed FBUS device is disabled until the PTEXECST parameter is set to ACTIVE.

Suppression of Process Alarms

The FBIOP does not generate return-to-normal events for process alarms that originated from the function blocks in the failed FBUS device when the PECM PTEXECST parameter is set to INACTIVE. To clear out the process alarms, the user must also set the PTEXECST parameter to INACTIVE for each FBCM in the failed FBUS device. When the FBCMs are inactivated, the process alarms associated with the inactivated FBCMs are removed from the Alarm Summary Displays.

Alarms in User Applications

Suppression of Device Alarms and Process Alarms During a Device Failure

6. Device Management in User Applications

6.1 System Management

System Management VFD

A System Management VFD represents a fieldbus device within the physical device and is an interoperable network view of a device application and its communication with another device or the TPS system. Each VFD specifies the communications characteristics of its device application and the FMS services used to access it.

TPS Builder VFD Templates

TPS Builder has a library which contains templates for manufacturer's VFDs.

System Management Key Features

The key features of system management operation:

- Provide system application clock time synchronization.
- Provide scheduling of function blocks.
- Provides Physical Tag and Address assignment of the device to the link.
- Provides a device identify service.
- Provides a find tag service.

System Application Clock Time Synchronization

Clock synchronization provides the capability for devices to time stamp data (events and alarms when they occur).

6.2 Device Identification

Each fieldbus device on the network is uniquely identified by:

- Device ID which is set by the manufacturer to uniquely identify the device.
- Device Name (Tag) — set by the user to identify operation.
- Device Address —a unique numerical address on the fieldbus segment (link). Address may be set automatically by system management, under Device Matching Direction from TPS Builder.

6.3 Scheduling a Control Strategy

Function Block Scheduling

The SMIB contains a schedule, called the function block schedule, that indicates the time that the device's function blocks are to execute.

System Management schedules the start of each function block relative to the execution period of the device. The execution period represents one complete cycle of the function block schedule in a device.

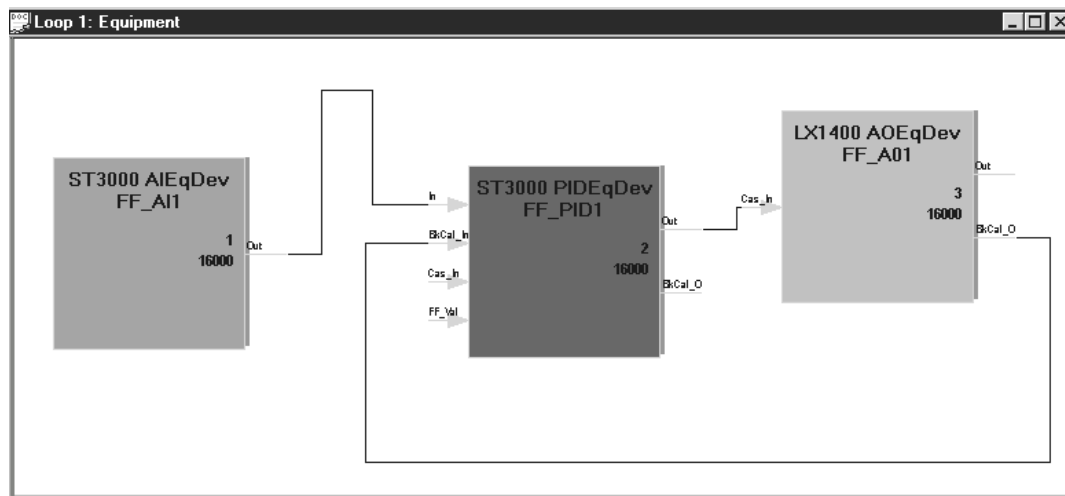
Link Active Schedule

Using the TPS Builder configuration software, the device's function block schedule can be pre-configured.

Control loops are configured on a link. A control loop consists of one to several function blocks that run in a designated order for a specified loop duration period.

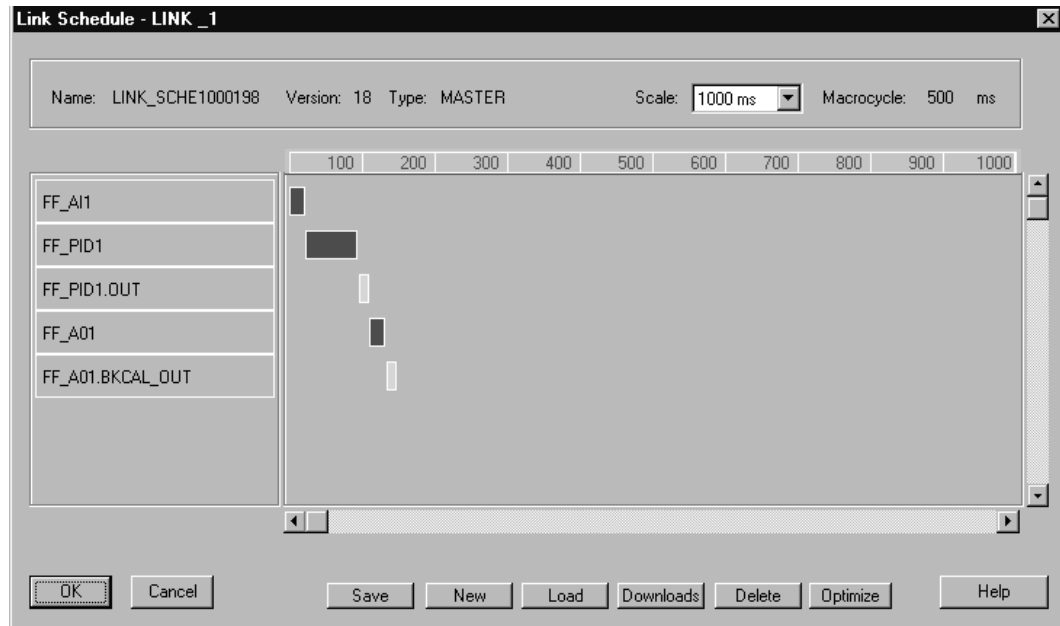
The following figure shows a typical control loop from TPS Builder's Equipment View.

Loop 1 Example



The following figure shows a control loop schedule from TPS Builder.

LAS Scheduling for Loop 1



Network Management

Network management is completely invisible to the user because the IOP interface and TPS Builder take care of all the network management functions.

7. Fieldbus IOP

7.1 Personality and Point Types

Fieldbus IOP Personality

The Fieldbus IOP is unique in that it comes from the factory with a loaded personality, and it must be reloaded from time-to-time as new function blocks are released by FOUNDATION Fieldbus and software is updated.

A reload of the personality is accomplished through the HPM Status Display, as shown later in the discussion regarding Status Displays.

HPM Fieldbus Point Types and Slot Numbers

The TPS system interfaces with the FOUNDATION Fieldbus system through a special IOP in the HPM. This IOP card can reside in any slot not used by the HPMM (two cards). The IOP functions as a “router” for the TPS Builder configuration functions. This routing is generally invisible to the user and is known to the TPS as the Blind Record Data Pipe, because the networks over which the data and messages are transferred do not understand the messages; they merely transfer them.

To complete the software interface, a new TPS point type, the Control Module point type was developed. It has three sub-types: the Segment Entity Control Module (SECM), the Physical Entity Control Module (PECM), and Function Block Control Module (FBCM). These point types serve as agents for the H1 segment, field device, and function/transducer blocks respectively.

Software Slot Assignment

For each fieldbus IOP, 128 software slots are available.

7.2 SECM Point Type

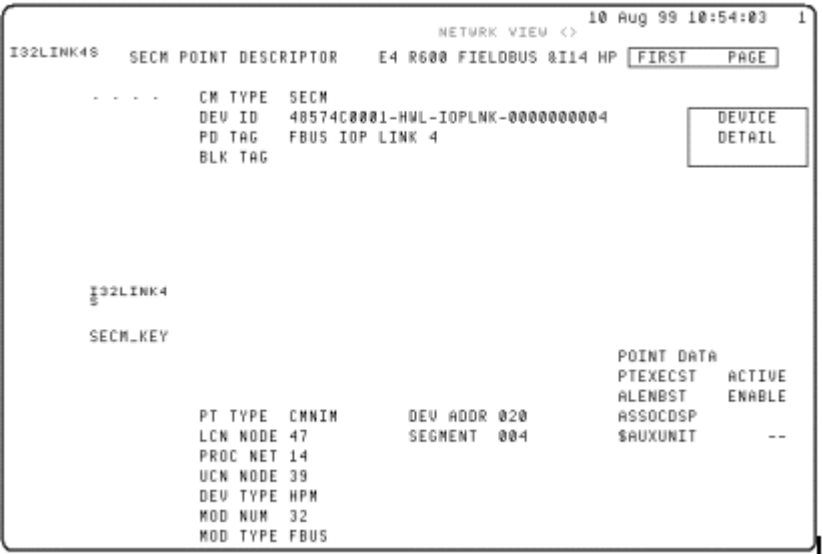
Segment Entity Control Module

Each Segment Entity Control Module (SECM, pronounced “sec’em”) represents a fieldbus network on one of the four H1 segments of the IOP. Network operational characteristics such as the active LAS are encapsulated by the SECM, as is the actual segment population and segment parameter configuration (that is, it indicates the device addresses present on the segment and the device type — IOP, Field Device, Visitor, Default_Address — of the connected devices).

A fieldbus link is associated with a fieldbus IOP by using the TPS Builder Parameter Configuration Form.

The following figure displays atypical SECM Point Detail Display.

SECM Point Detail Display



Command SECM shutdown from Device Detail

You can shutdown a segment from the Device Detail display, by selecting the **COMMAND** target, as shown in the following figure.

Command targets on SECM Device Detail Display

04 Aug 99 14:02:13 7

IOP7LINK15
LINK POINT
NETURK VIEW <> A1 APM 241

LINKCMD
RESTART
SHUTDOWN
NULL

PRIMARY LINK MASTER ON
PD TAG FBUS IOP LINK 0
PT TYPE SECM

FIRST PAGE

REFRESH

SEGMENT STAT STK LOED
PTEKECST INACTIVE
COMMAND NONE

FB NETWORK 1
SEGMENT 01
SEG NAME IOP7LINK15
DEV ADDR 20

PHY DEV MODEL SERIAL NR 48574C0001-HUL-IOPLNK-0000000000
MANUFACTURER DEV MODEL NR R600 FBUS IOP
MANUFACTURER NAME HONEYWELL
REVISION NUMBER REV 0.00

LINK I/F ERRORS

ENTER

UCN 1

IOM 7

Fbus Seg

Fbus Net

Block

HPM 19

SLOT 120

Status

Statistics

Detail

FTWARS63.06

SECM Livelist Bitmap

The SECM Status Page contains the livelist bitmap, which directly corresponds to the Fieldbus Segment Status. The following illustrates a typical segment status display.

Segment Status Display Page 1

09 Aug 99 17:43:51 1

SEGMENT: I32LINK4S
SECM POINT DESCRIPTOR
OK

01 DEV ADDR 22 DEV OK 5 48574C0002-HWL-ST3000-DEFCBA9000	02 DEV ADDR 26 OFFNET 0 48574C0002-HWL-ST3000-DEF8447000
03 DEV ADDR 29 DEV OK ALX 464C530201-VAL-LX1400-00DEF661D0	04 DEV ADDR 28 DEV OK LX82 464C530201-VAL-LX1400-00DEF78C10
05 DEV ADDR 24 DEV OK ST75 48574C0002-HWL-ST3000-DEFF28F000	06 DEV ADDR 30 DEV OK ST74 48574C0002-HWL-ST3000-DEFF04E000
07 DEV ADDR 21 DEV OK LX71 464C530201-VAL-LX1400-00DEFB710	08 DEV ADDR 23 DEV OK ST71 48574C0002-HWL-ST3000-DEFB805000
09 DEV ADDR 25 DEV OK ST72 48574C0002-HWL-ST3000-DEFBF9D000	10 DEV ADDR 27 DEV OK ST73 48574C0002-HWL-ST3000-DEF89E7000

UCN 14

IOM 32

FTWABS60.06

Prev
Seg

Next
Seg

Fbus Net
Statistics

Device
Detail

17767

7.3 PECM Point Type

Physical Entity Control Module

Each **Physical Entity Control Module** (PECM, pronounced “peck’em”) represents one fieldbus network physical device and encapsulates the data it contains which characterize its network operational behavior.

PECM Point Detail Displays

A PECM point first page is shown in the following figure. A PECM has no group display.

PECM Point Display

10 Aug 99 14:41:07 1

NETWORK VIEW <>

RST3K1P
REAL ST3000 DEVICE
E4 R600 FIELDBUS 0I14 HP

FIRST PAGE

CM TYPE PECM

BLK TAG

VEN NAME Honeywell

MODEL ST3000FF

VFD REV Rev 3.00

DEVICE
DETAIL

RST3K1P

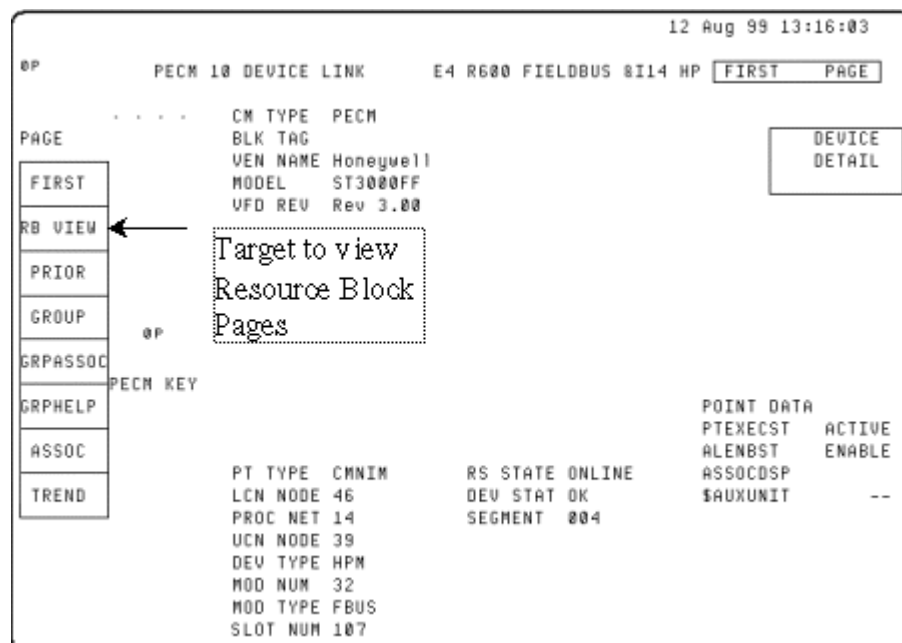
RST3KEY

		POINT DATA	
PT TYPE	CMNIM	RS STATE	ONLINE
LCN NODE	47	DEV STAT	OK
PROC NET	14	SEGMENT	003
UCN NODE	39		
DEV TYPE	HPM		
MOD NUM	32		
MOD TYPE	FBUS		
SLOT NUM	117		

Targets to PECM Pages

The targets on the various PECM pages give the user access to other displays related to this point.

PECM with Targets to View Resource Block Pages



PECM Resource Block View

The first page, subpage 1 of the Resource Block is shown in the following figure. The resource block has the following views in addition to the display in the following figure:

- Operator Static
- All Dynamic
- Other Static

Resource Block — Operator Dynamic, Subpage 1

12 Aug 99 13:03:55 1

OP
PECM 10 DEVICE LINK E4 R600 FIELDBUS 8I14 HP
RB VIEW PAGE 1

Operator Dynamic Data - ST3000 Resource Block SUBPAGE 1

PARAMETER	INDEX	VALUE
ST_Rev	129	
Mode_Blk	TARGET	: Auto
	ACTUAL	: Auto
	PERMITTED:	00S, Auto
	NORMAL	: Auto
Block_Err		Other, SimulationActive
RS_State		Online
Free_Time		78.4610
Fault_State		Clear

7.4 FBCM Point Type

Function Block Control Module

A **Function Block Control Module (FBCM)** represents a portion of a control strategy. An FBCM point contains one function block and/or one transducer block. Function blocks within the FBCM are linked using TPS Builder.

The types of FBCMs which may be built are:

- a “basic” control FBCM which encapsulates a single FBAP process function block.
- a “basic” I/O FBCM which encapsulates a single FBAP process I/O function block and a single transducer block.
- a “basic” transducer FBCM which encapsulates a single FBAP transducer block.

Block Pages

Block (VIEW) pages apply to function blocks and transducer Blocks assigned to a FBCM and resource blocks assigned to a PECM.

Following is an example of an operator dynamic data page for the fieldbus AI.

FB VIEW Analog Input Operator Dynamic Data

			10 Aug 99 11:16:18	1
			NETWORK VIEW <>	
AI-ST-CMTRF	AI ST3000	E4 R600 FIELDBUS &I14 HP	FB VIEW PAGE	1
Operator Dynamic Data - ST3000 AI Block			SUBPAGE	1
PARAMETER	INDEX	VALUE		
ST_Rev	286			
Mode_Blk	TARGET	: Auto		
	ACTUAL	: Auto		
	PERMITTED:	OOS, Man, Auto		
	NORMAL	: Auto		
Block_Err	SimulationActive			
PV	89.7771	* CRIT ALARM		
Out	89.7771	* CRIT ALARM		
Field_Val	89.7771	* CRIT ALARM		

7.5 Parameter Mapping and Projection

Access to Parameters

In order for the detail, group, alarm, and network displays to gain access FF parameter, process data such as setpoints, process variables, mode, etc. are brought into traditional TPS parameters wherever possible. Parameter projection and parameter mapping concern the ability to view the value of a fieldbus parameter in the corresponding TPSN CM strategy.

Mapped Parameters

Parameter mapping is used when there is a one-to-many relationship between a fieldbus parameter and a TPN parameter. This is normally used to support the fieldbus structured parameter.

Manufacturer specific parameters may be added to the TPS Builder repository during processing of a Device Description. If required, these parameters may be mapped so that they can be accessed in TPN.

Projected Parameters

Parameter projection is used when there is a one-to-one relationship between a fieldbus parameter and a TPN parameter.

Structured Parameters

A structured parameter is a parameter that is composed of multiple elements or variables. These elements or variables are modeled as sub-parameters and must be of the FF basic variable types. These element parameters are usually read or written as one unit.

7.6 Peer-to-Peer Connections to Field Device Parameters

Field Interconnections

For FF-compliant devices, such as transmitters which contain a function block (in this case, an AI) to make its primary variable, the OUT parameter available to other blocks, an inter-device connection is needed. The most efficient way for TPS to collect these values on a periodic and timely basis is for the LAS to call for the desired parameter to be published on the wire periodically. The Virtual Communications Relationship (VCR) is the FF term for the named set of parameters that make possible inter-device connections.

Control Interconnections

The methods of control connections between TPS and FF devices differ in that the fieldbus IOP supports control downstream only. That is, it controls downstream function blocks but does not permit control by an upstream block that resides on the fieldbus.

The fieldbus IOP can communicate to a downstream block using either of two methods: publication (publish/subscribe) or writing messages (client/server). The publication technique permits the IOP to appear to the segment to be another FF device. The published IOP output is subscribed to by the receiving device-resident block such as an analog output (AO) or PID, usually by its CAS_IN (input) parameter connection. This connection is generally utilized when the downstream control block is in Cas (cascade) mode.

Remote Out or Direct Digital Control

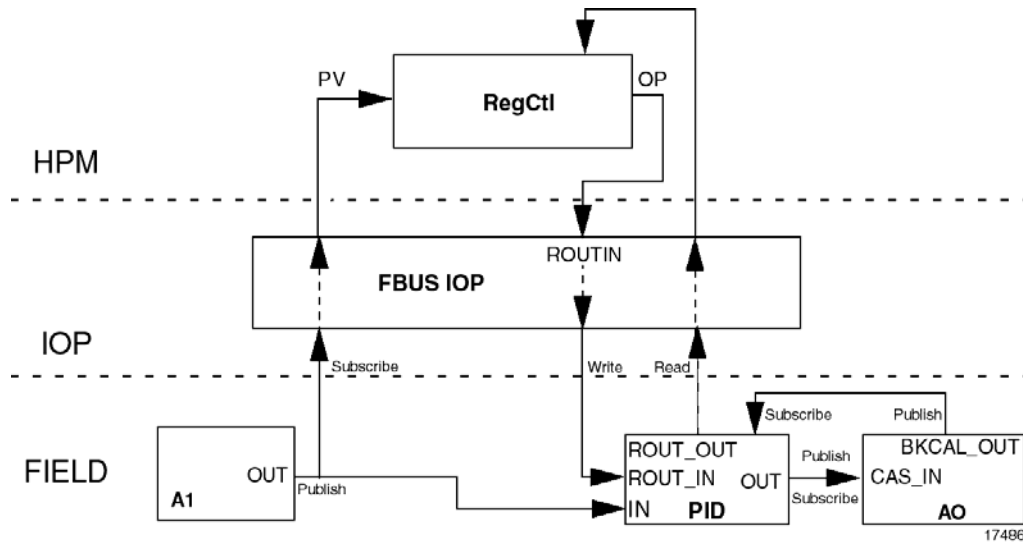
The client-server technique permits the IOP to appear to the FF segment to be a “computing device.”

The client/server technique permits the IOP to act in what is sometimes called (Direct digital control) DDC mode (Remote out in FF terms).

Regulatory Control in HPM, AI, Backup PID and AO in Field

Fieldbus IOP

Peer-to-Peer Connections to Field Device Parameters



Consistent Control Relationship

The relationship between function blocks in FF is consistent: downstream action with upstream feedback. The relationship of the fieldbus IOP to an FF AO is the same as the relationship of the fieldbus IOP to an FF PID.

Control Connections through HPM

Within the fieldbus IOP, devices on one segment cannot communicate with devices on another, except through connections to control algorithms in the HPM.

7.7 Parameter Support in Displays

Schematics

Standard and custom displays are provided with process and maintenance through the fieldbus IOP.

Fieldbus Additions to Conventional Displays

Standard operating schematics and displays have been modified to include field bus data. The following table summarizes these changes:

Display	Modifications
HPM Status Display	Add support for FB IOP
IOM Detail Display	Add navigation targets
IOP Slot Summary	Add support for displaying CM in FB IOP slots
Detail Display	A new point type for fieldbus has been added
Group Displays	Includes the new fieldbus point type
Organizational Display	Adds the CM subtype

New Displays for Fieldbus

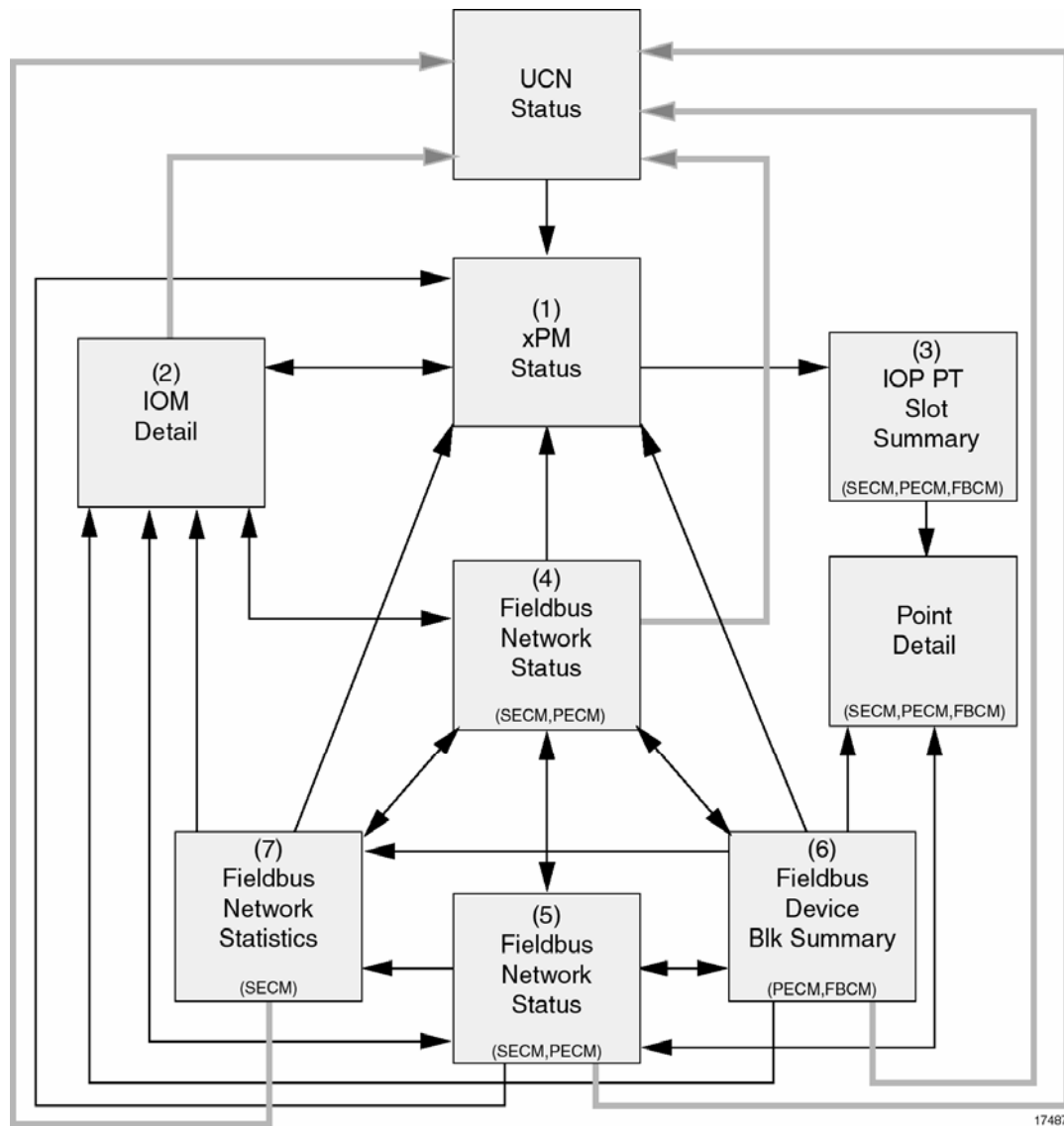
New displays are added to support fieldbus.

Display	Content
Fieldbus Network Status	ID/Status of SECMs, PECMs on a FB IOP
Fieldbus Device Detail	Status/detailed data for a PECM or SECM
Fieldbus Device Slot Summary	ID/Values of all FBCMs belonging to a PECM
Fieldbus Network Statistics	Statistical data for FB devices on a FB IOP

Calling Fieldbus Network Displays

The calling tree for fieldbus network displays is shown in the following figure.

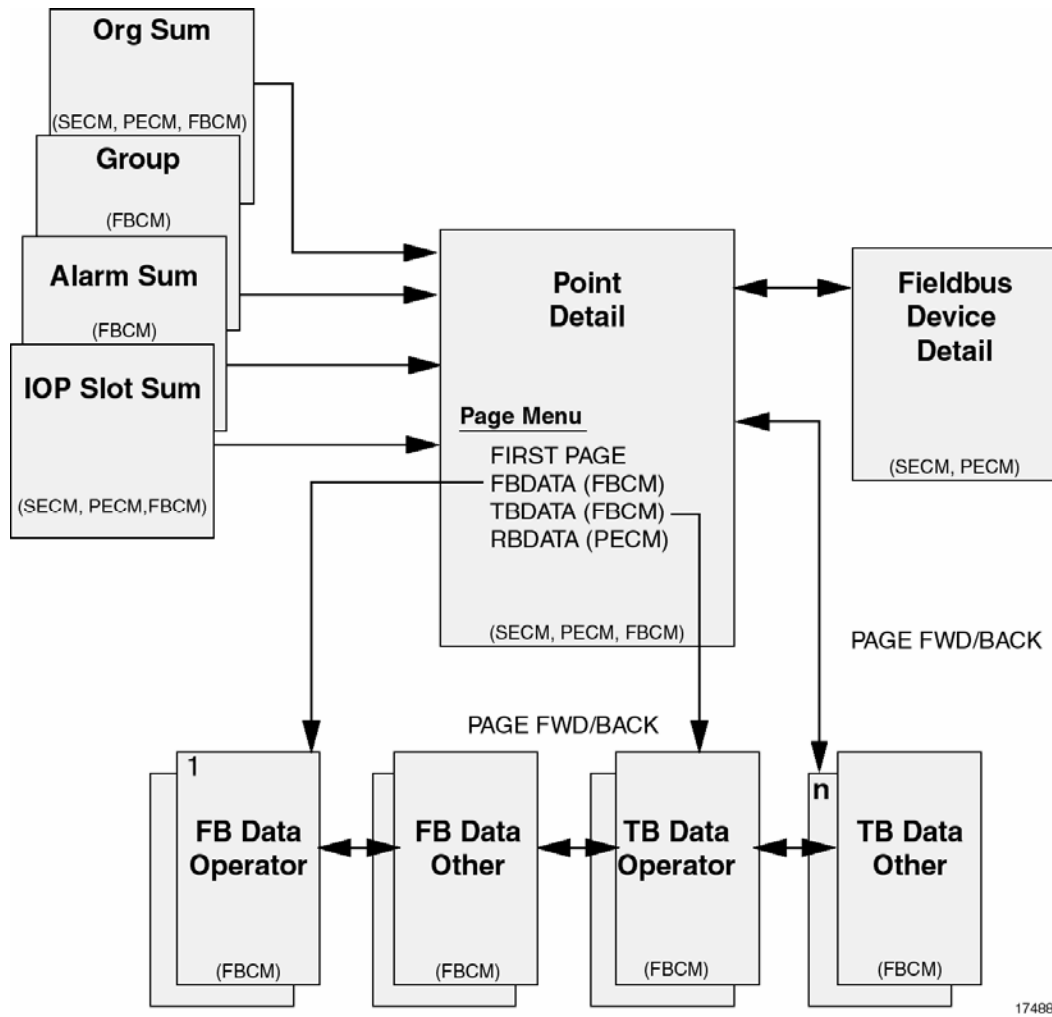
Calling Tree Fieldbus Displays



Calling Fieldbus Control Module Displays

The calling tree for fieldbus control module displays is shown in the following figure.

Calling Tree CM Point Detail Displays



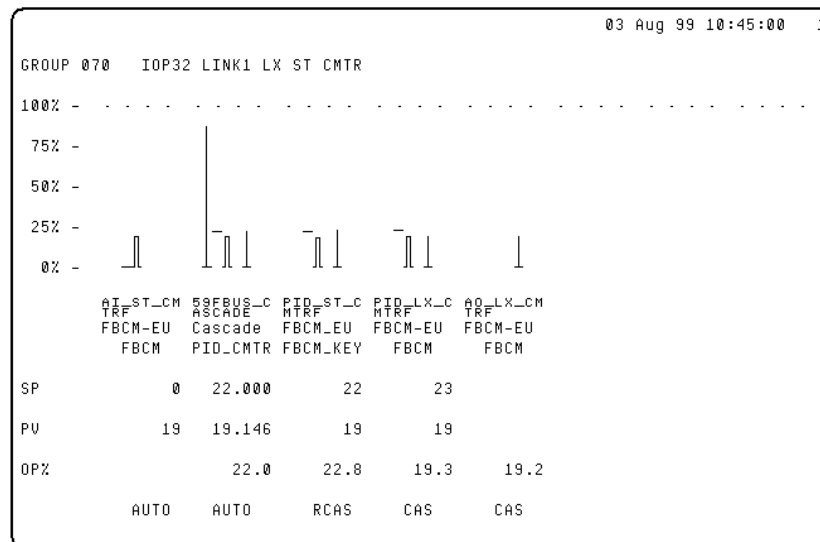
CL Restrictions

Control Strategies using the proprietary CL language can access the fieldbus parameters that are mapped or projected to TPS parameters with certain restrictions as described in Scheduling a Control Strategy.

Group Display

Group Displays are built using the Group Display. The group display is able to hold points with the current available types plus fieldbus CM points with a control module type (CMTYPE) of FBCM is an example of fieldbus group display.

Example of Group Display



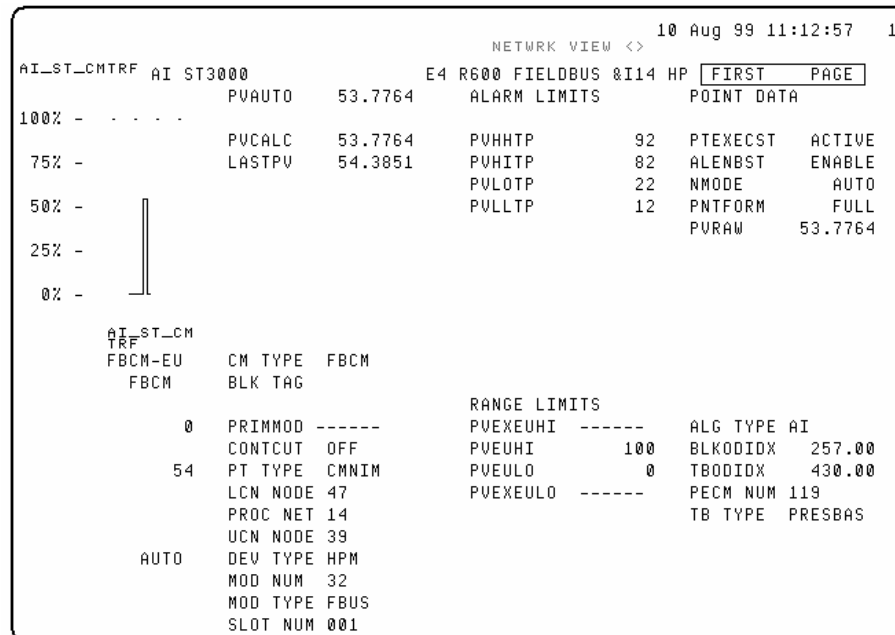
Metadata Based Detail Display

As the TPS builder completes the configuration (or modification) of a fieldbus network for a UCN, it writes a new Detail Display Metadata file to the HM. This metadata is used primarily for display purposes.

Detail Display

The fieldbus point detail display is similar to other TPN displays, as shown in the following figure.

Example of AI Point Detail Display



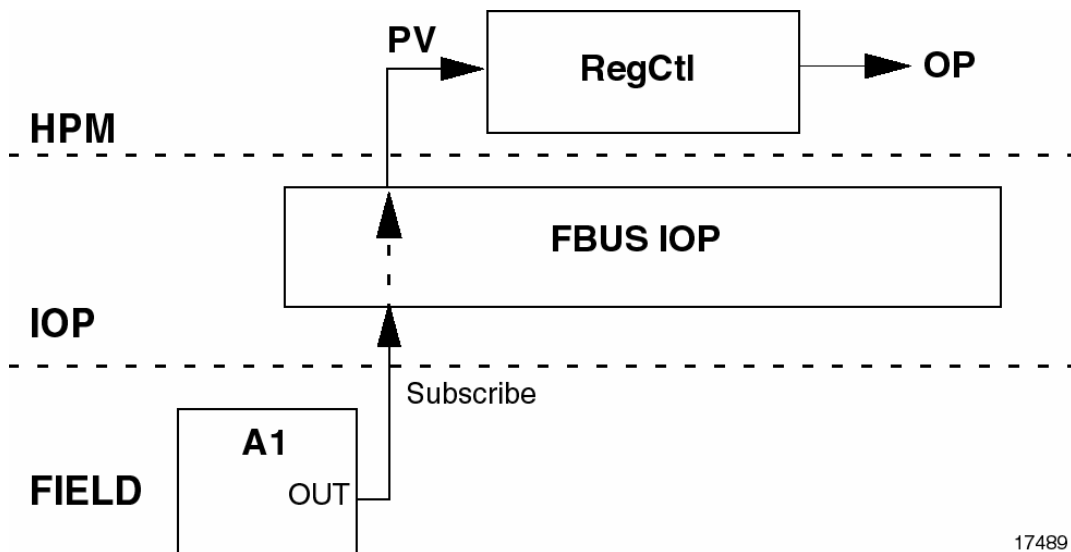
8. HPM and Fieldbus Control Considerations

8.1 HPM Control Loss, Failure, and IOP Restart

HPM Control Loss

Control algorithms can reside in the HPM and/or as function blocks in the device. Fieldbus H1 segments can be configured to have power supplies and battery backup independent of power to the HPM; therefore the bus segment and the devices could retain power if power was lost or interrupted in the HPM.

Input in Fieldbus Device, Regulatory Control and Output in HPM



17489

HPM Failure

Upon detection of an HPM failure a system alarm will be generated. Restart is through the HPM status display. As the IOP is HPM resident, no point processing takes place. Processing is stopped and outputs are held. Slot processing takes place only when the HPMM box state is RUN.

Fieldbus IOP Restart

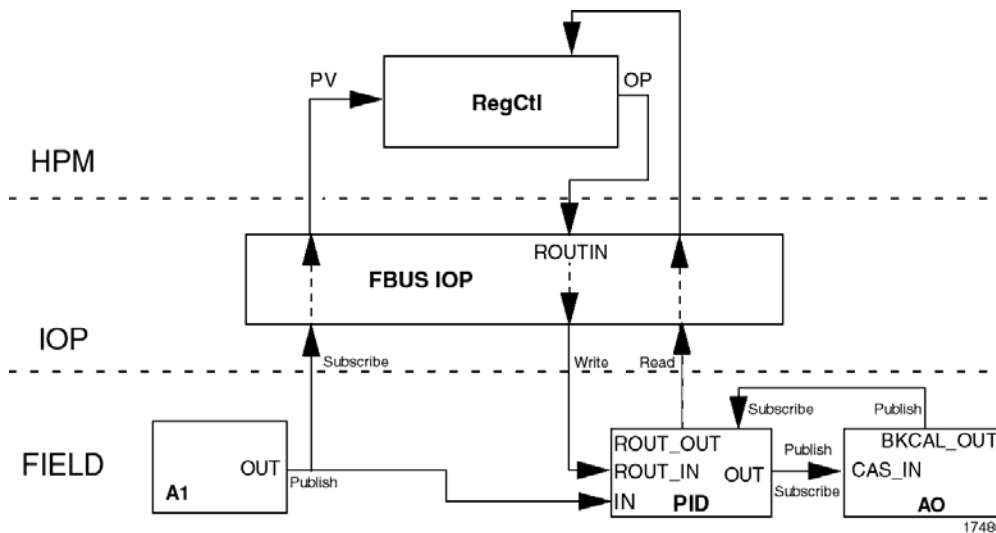
The IOP is the interface for TPS and the fieldbus network. It contains the scheduling mechanism (LAS) and is responsible for collecting data and mapping or projecting the

FF parameters to TPS parameters. Under normal operating conditions the IOP serves as the Link Master of the fieldbus segment. However, the IOP resident Link Master may have another field device, such as the Honeywell ST3000, which is capable of acting as the Link Master.

8.2 HPM Failover to Fieldbus Control

If a control strategy is configured in the HPM, it can also be configured to the fieldbus network as a backup, as shown in the following figure. This configuration can be done providing the fieldbus network and the HPM are not sharing common power and backup battery sources or the problem is not in the power source.

RegCtl PID in HPM, AI, AO and Backup PID in Fieldbus



9. Displaying HPM Fieldbus Points

9.1 Schematic Parameter Access

Accessibility defined in PRD

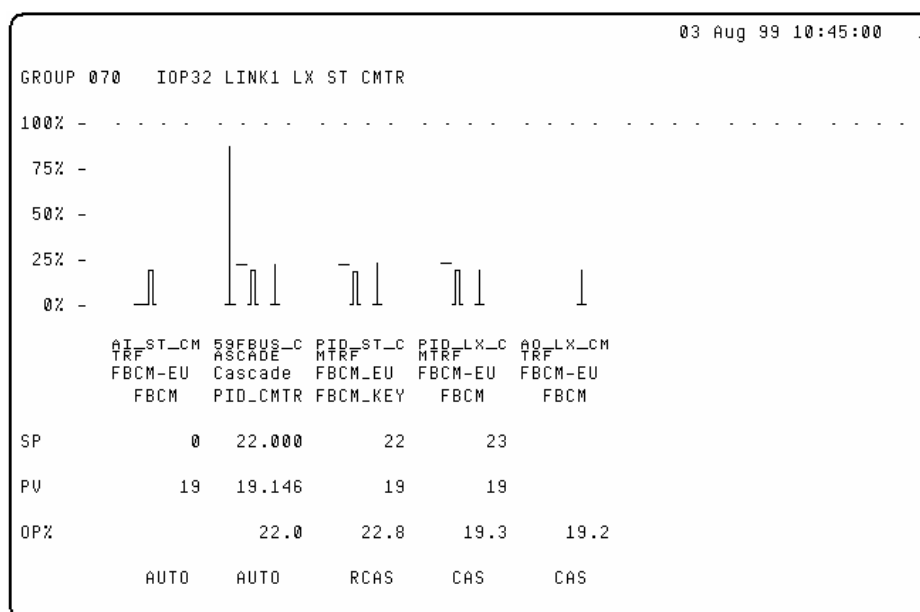
Custom schematics can access and parameter and the value for that parameter that is shown to be accessible in the Parameter Reference Dictionary (PRD). Some parameters are structured, in that they contain more than just the value; they may also contain a status. Refer to the PRD for the projection and mapping of the status byte. However, some of the fieldbus parameters are “view only” and are not accessible; these are not parameters usually used on custom schematics.

9.2 Group and Detail Display Behavior

Group Display Behavior

SECM and PECM points can not be added to a group display, but any other legitimate point type, such as an HPM RegCtl point can be added to a group containing FBCM points. The general format of a Group Display is shown in following figure.

Group Display Including Fieldbus Points



Fieldbus points are similar to other point displays, except for the Mode parameter, which is a composite of the fieldbus Actual and Target Modes. Parameters which are changeable can be selected and changed from the Group Displays.

Point Type Detail Displays

Detail displays are provided for the following point types:

- FBCM — in addition to the displays shown in the following figures, also see detail and block pages in the discussion regarding the FBCM Detail Display.
- PECM (See displays in PECM Point Type discussion.)
- SECM (See displays in SECM Point Type discussion.)

Status Byte Indication

Value fields also display the FF sub-index names if applicable for the data. FF Data types DS-65 (value and status, floating point structure), DS-66 (value and status of discrete

value parameters) and DS-67 (value and status bitstring) have a status byte that is represented on displays as a 1, 3, or 16 character string following the value.

The status is composed of three parts:

1. A quality indication
2. A substatus
3. Limit indication

9.3 Display Mode Handling

Mode Management

Fieldbus has eight modes; these operating modes differ somewhat from other TPS operating modes in that FF has the concept of mode attributes: Normal, Target, Permitted and Actual.

Rotary and Switch Models

For reasons of compatibility with various manufacturers' equipment, the FF specifications provide for two "models" of operation: (1) the "rotary switch" model and (2) the "toggle switch" model. Honeywell has traditionally supported the rotary switch model. For consistent user access, TPN permits entries for the rotary model (one exception noted below). However, since third party devices can access FF devices, if the toggle model forms (multiple targets requested) are ever encountered, the target mode which is dominant will be displayed.

The "rotary switch" model means that only one mode can be requested at a time (for example, OOS, Man, Auto, Cas, RCas, ROut). A characteristic of that model is that there is no memory of previous target modes. This is supported by TPN.

The "toggle switch" model means that more than one mode may be requested at a time (for example, Manual Override of Cascade, Manual Override of Remote Cascade, etc.).

Configuration Options

Since this is a configuration option, as opposed to an operator selection option, an FBCM-resident parameter, RETCAS, indicates whether the Cas target should be set (or not) automatically (by the FBUS IOP) every time the operator enters either RCas or ROut as the requested target mode or either of those modes are the “normal” mode

Additionally, at configuration time, the user can choose to change the description of ROut or RCas modes to be Cas or BCas on displays using (MODE and MODEBKLC parameters) such as the group and detail displays. An FBCM-resident parameter, MODEDOPT, controls the mode display options as follows:

9.4 Status Displays

Displays overview

The following displays are used to determine Fieldbus Network Status

- UCN Status Display
- HPM Status Display
- IOM Detail Display
- Fieldbus Network Status Display
- Fieldbus Device Block Summary
- Fieldbus Network Statistics
- IOP Point Slot Summary
- Fieldbus Device Detail

IOM Detail Display

Following is an example of a typical IO Module display for a fieldbus IOP.

IO Module Display for a Fieldbus IOP

MAKE SELECTION04 May 99 01:48:10 1

FBUSNon-redundant SF

UCN 14HPM 39MODULE 37 PHYS ADDR 231 FILE 7 CARD 8

STS SOFTFAIL ACTUAL TYPE FBUSFTA Connect CONN_A

Status:

HARDWARE REV 50 FIRMWARE REV 3.9 LAST HARD FAILURE SPARE053

DESCRIPTION: FBUS H1

ACTIVE CABLE CHANNELB CHN A ERR : 0 CHN A SIL : 0

LAST COMM ERR NONE CHN B ERR : 0 CHN B SIL : 0

RESET MODULE
COMM ERRORS

FTA PRESENT

NO STANDBY MANUAL

COLD POWERUP LAST EXECUTED NO SWITCHED ACTIVE

Box Soft Failures

Slot Soft Failures

Slots with Soft Failures

FB NET 1 SEG 1

FB NET 2 SEG 1

SEG

SEG

Refresh Display

PI Load Info

Fbus Seg Status

Device Detail

HPM/Fieldbus Checkpointing

The data owner for fieldbus points is the NIM and the checkpoint file is stored on the History Module. Fieldbus IOP point data save/restore is accessed through the System Status Display. Select targets NIM and NTWK/HWY STATUS. This brings up the UCN Status display. Select the HPM in which the fieldbus IOP resides and pick the target DETAIL STATUS. This brings up the HPM Status display, from where the program and data save/restore can be invoked. For R6XX a new file type is added to the UCN checkpoints: FXXNNYYM.CP

FB IOP Personality Upgrade

The fieldbus IOP is loaded before shipment and must be reloaded as necessary when new releases are distributed. If it is necessary to reload this personality, use the HPM Status Display.

Following is are examples of a typical HPM status page and an IOP PI loading information page.

Displaying HPM Fieldbus Points Status Displays

HPM Status Display with Fieldbus IOP Personality Reload

LOAD PROGRAM	LOAD MODULE	RESTORE ALL	RESTORE ALL	SAVE ALL	SIMULATOR LOAD	CANCEL	ENTER
RESTORE STATES	CONTROL	AUTO CHECKPT	TOL CABLE COMMANDS	RUN STATES	SLOT SUMMARY	DETAIL STATUS	
RESTORE	LOAD/SAVE						

01 DISOE	02 OK	03 BK\OK	04 OK \BKP	05 HLAI	06 OK	07 OK	08 OK
09 HLAI	10 OK	11 OK	12 OK	13 OK	14 OK	15 OK	16
17 DI	18 OK	19 OK	20 DISOE	21 OK	22 OK	23	24
25	26	27	28	29 FBUS	30 00-32	31 FBUS	32 FBUS
33 00-16	34 LJMUX	35 HLAI	36 00-16	37 FBUS	38 DI	39 DISOE	40 DISOE
OK	2F	OK	OK \BKP	2F	OK \BKP	OK	OK

HPM AUTO CHECKPT: ENABLE
 IOL PERIODIC SWAP: ENABLE
 PFI FILE TO BE LOADED:
 WRITE LOCKOUT : OFF

HPM 38 STATUS\UIN 14
 HPM 40 2 HPM 38 P
 BACKUP
 OK

HPM CONTROL STATE : FULL
 A\B HPM 40 2 HPM 38 : A\B

1 01 00 00 10:27:34

NETWORK VIEW <>

ENTER FILE NAME

IOP PI Loading Information Display

04 May 99 01:40:49 1

IOP PI LOADING INFORMATION DISPLAY

UCN 14 HPM 39 IOM 37

PERSONALITY LOCK: C000 0000 0000 0001
CURRENT PI FILE NAME: fbio6039
NEXT PI FILE NAME:

STATUS: SUCCESS LOADING MODE: 0 PACKET NUMBER: 4584

PERSONALITY INFORMATION
IDENTIFIER: Field Bus IOP
KEY: 0000 0000 0000 0001
VERSION: 3
REVISION: 9

Displaying HPM Fieldbus Points Status Displays

Segment Status Display Pages

The segment status display shows the devices configured on the network, their device addresses and their individual states, as shown in the following figure.

Fieldbus Segment Status Display

09 Aug 99 17:43:51 1

SEGMENT: I32LINK4S
SECM POINT DESCRIPTOR OK

REFRESH

PAGE 1

01 DEV ADDR 22 DEV OK 5 48574C0002-HWL-ST3000-DEFCBA9000	02 DEV ADDR 26 OFFNET 0 48574C0002-HWL-ST3000-DEF8447000
03 DEV ADDR 29 DEV OK ALX 464C530201-VAL-LX1400-00DEF661D0	04 DEV ADDR 28 DEV OK LX82 464C530201-VAL-LX1400-00DEF78C10
05 DEV ADDR 24 DEV OK ST75 48574C0002-HWL-ST3000-DEFF2BF000	06 DEV ADDR 30 DEV OK ST74 48574C0002-HWL-ST3000-DEFF04E000
07 DEV ADDR 21 DEV OK LX71 464C530201-VAL-LX1400-00DEFBB710	08 DEV ADDR 23 DEV OK ST71 48574C0002-HWL-ST3000-DEFB805000
09 DEV ADDR 25 DEV OK ST72 48574C0002-HWL-ST3000-DEFBF9D000	10 DEV ADDR 27 DEV OK ST73 48574C0002-HWL-ST3000-DEF89E7000

UCN 14

IOM 32

Prev
Seg

Next
Seg

Fbus Net
Statistics

Device
Detail

HPM 39

FTWABS60.06

Network Statistics Display

The Fieldbus Network Statistics shows the number of segments (networks or links) configured on an individual FBIOP, the UCN, HPM, and IOM module numbers. Up to four segments can be configured as shown in the following figure.

Fieldbus Network Statistics

12 Aug 99 13:29:07 1

UCN 14 HPM 39 IOM 32

REFRESH

FB NET 1	SEG	1
FB NET 2	SEG	1
FB NET 3	SEG	1
FB NET 4	SEG	1

This page cannot be filled in any further since statistical parameters have not yet been defined.

Fbus Seg
Status

Displaying HPM Fieldbus Points

Status Displays

10. Display Indications of FF Status Byte

10.1 Status Byte and Data Structure Overview

Status Byte Information

Every variable and some contained fieldbus values have a status byte as an element of the structure; therefore, every process variable has this property.

Data Structures

Every connectable input and output supports the status byte.

Status is used to indicate

- Data Quality (Usability)
- Data Failure Cause
- Data Degradation Cause
- Limit Conditions (Upward & Downward)
- Cascade Control Initialization, Rejection (Upstream Block Class Indication)
- Fault-State Initiation, Indication (Upstream Block Class Indication)
- Local-Override Indication
- Worst-Case Alarm Indication (Upstream Block Class Indication)

Coding Looks Like:

Quality		Sub-status				Limits	
msb							lsb

Limits:

- 0 = No Limits
- 1 = Low Limit
- 2 = High Limit
- 3 = Constant

Quality:

- 0 = BAD Data Quality
- 1 = UNCERTAIN Data Quality
- 2 = *GOOD (Non-Cascade) Data Quality [for example, used by AI's OUT, DI's OUT_D]
- 3 = GOOD (Cascade) Data Quality [for example, used by PID's OUT, BKCAL_OUT]

Sub-Status if Quality is BAD:

- 0 = Non-Specific
- 1 = Configuration Error
- 2 = Not Connected
- 3 = Device Failure
- 4 = Sensor Failure
- 5 = No Communication, with Last Usable Value
- 6 = No Communication, with no Last Usable Value
- 7 = Out-Of-Service

Sub-Status if Quality is UNCERTAIN:

- 0 = Non-Specific
- 1 = Last Usable Value
- 2 = *Substitute
- 3 = *Initial Value
- 4 = Sensor Conversion Not Accurate
- 5 = Engineering Unit Range Violation
- 6 = Sub-Normal

Sub-Status if Quality is †GOOD (Non-Cascade):

- 0 = *Non-Specific
- 1 = *Active Block Alarm
- 2 = *Active Advisory Alarm
- 3 = *Active Critical Alarm
- 4 = *Unacknowledged Block Alarm
- 5 = *Unacknowledged Advisory Alarm
- 6 = *Unacknowledged Critical Alarm

Sub-Status if Quality is GOOD (Cascade):

- 0 = Non-Specific
- 1 = *Initialization Acknowledge (IA)
- 2 = *Initialization Request (IR)
- 3 = *Not Invited (NI)
- 4 = *Not Selected (NS)
- 5 = *Do Not Select (DNS)
- 6 = Local Override (LO)
- 7 = *Fault-State Active (FSA)
- 8 = *Initiate Fault-State (IFS)

* Not used by OPC (the Object Linking and Embedding for Process Control)
version 1.0

Display Indications of FF Status Byte
Status Byte and Data Structure Overview

Reader Comments

Fax Transmittal		Fax No.: (602) 313-4212
Reader Comments		
To:	Information Development	
From:	Name	Date:
	Title:	
	Company:	
	Address:	
	City:	State: Zip:
	Telephone:	Fax:
TPS Fieldbus Overview, FB27-630		
Release R630, 12/01		
Comments:		
You may also call 800-343-0228 (available in the 48 contiguous states) or write to:		
Honeywell Industrial Automation and Control 16404 North Black Canyon Highway Phoenix, AZ 85053		Knowledge Building Tools 

Honeywell

Honeywell
Industrial Automation and Control
16404 North Black Canyon Highway
Phoenix, AZ 85053